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Influence of varied electrode geometries on machining performance and surface topography in die-sinking EDM of Al2219 alloy

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ABSTRACT

Contributing to SDG 9 (Industry, Innovation, and Infrastructure), this study systematically investigates the influence of electrode geometry on machining performance and surface integrity during die-sinking electrical discharge machining (EDM) of blind holes in Al2219 alloy. Die-sinking EDM offers a viable solution, yet its application is often constrained by low material removal rates (MRR) and poor surface integrity, particularly in blind hole configurations where debris evacuation is challenging. Electrode geometries were designed and fabricated, including a conventional cylindrical design along with Straight radius cut, Groove shaped, Flute shaped, and Relief Angle 10° design. The baseline provided MRR values of 6.238 ± 0.022, 13.148 ± 0.127, and 21.603 ± 0.085 mm^3/min for Pulse 1, 2, and 3, respectively. The modified designs generally improved performance compared to baseline with D_{NC-6} delivering peak MRR at both low and high currents: 9.567 ± 0.045 mm^3/min at 3A and 36.431 ± 0.292 mm^3/min at 7A, while D_{NC-8} improved at 5A, achieving 34.712 ± 0.204 mm^3/min. Other factors including tool wear rate, machining time, surface roughness, and overcut were also examined. EDX line scans showed that carbon from dielectric decomposition remained confined to the recast layer, with no diffusion into the substrate, and XRD confirmed that no new crystalline phases formed. The results demonstrate that electrode design exerts a significant influence on all measured responses. The highest MRR improvement was achieved by D_{NC-8} at 5A, with a 164.0% increase (95% CI 160.9–167.1%, adjusted $p < 0.001$) over the conventional electrode.

1. Introduction

Electrical discharge machining (EDM) die-sinking is a non-conventional metal removal process used to cut difficult to machine materials and features in aerospace, automotive, and other die making applications. This process removes material with the application of controlled electrical discharges between electrode and workpiece submerged in dielectric fluid [1,2]. This process enables machining of hardened alloys, superalloys, and other composites. Among its various other applications die-sinking EDM remains the most widely used technique for the manufacturing of intricate and precision blind holes, a common requirement in automotive and aerospace sectors [3–5]. Blind holes are frequently required to secure locking pins of different mechanisms, sensor housings, or create oil channels inside different structural components. These features must withstand repeated vibration and thermal cycle during vehicle or aircraft operation, making their dimensional precision a quality control requirement for feature or component fabrication [6].

Effective flushing remains one of the most significant challenges in die-sinking EDM, particularly for blind holes and deep cavities [7]. Inadequate flushing can lead to secondary discharges, short circuits, higher machining times and poor surface quality of workpiece surface. This phenomenon is more noticeable in deep holes [8]. Research has shown that increasing flushing intensity to achieve higher Reynolds numbers enhances plasma flushing efficiency and improves material removal rate (MRR) [9]. Researchers also investigated the effects of various flushing conditions on deep hole drilling accuracy [10]. For higher aspect ratio holes, it was observed that through flushing was better than side flushing or its combination with through flushing [11]. Despite progress in this area, achieving improved flushing remains one of the key drivers of electrode geometry and design modifications.

Electrode geometry significantly influences performance parameters such as TWR, MRR and dimensional accuracy. Early studies investi-

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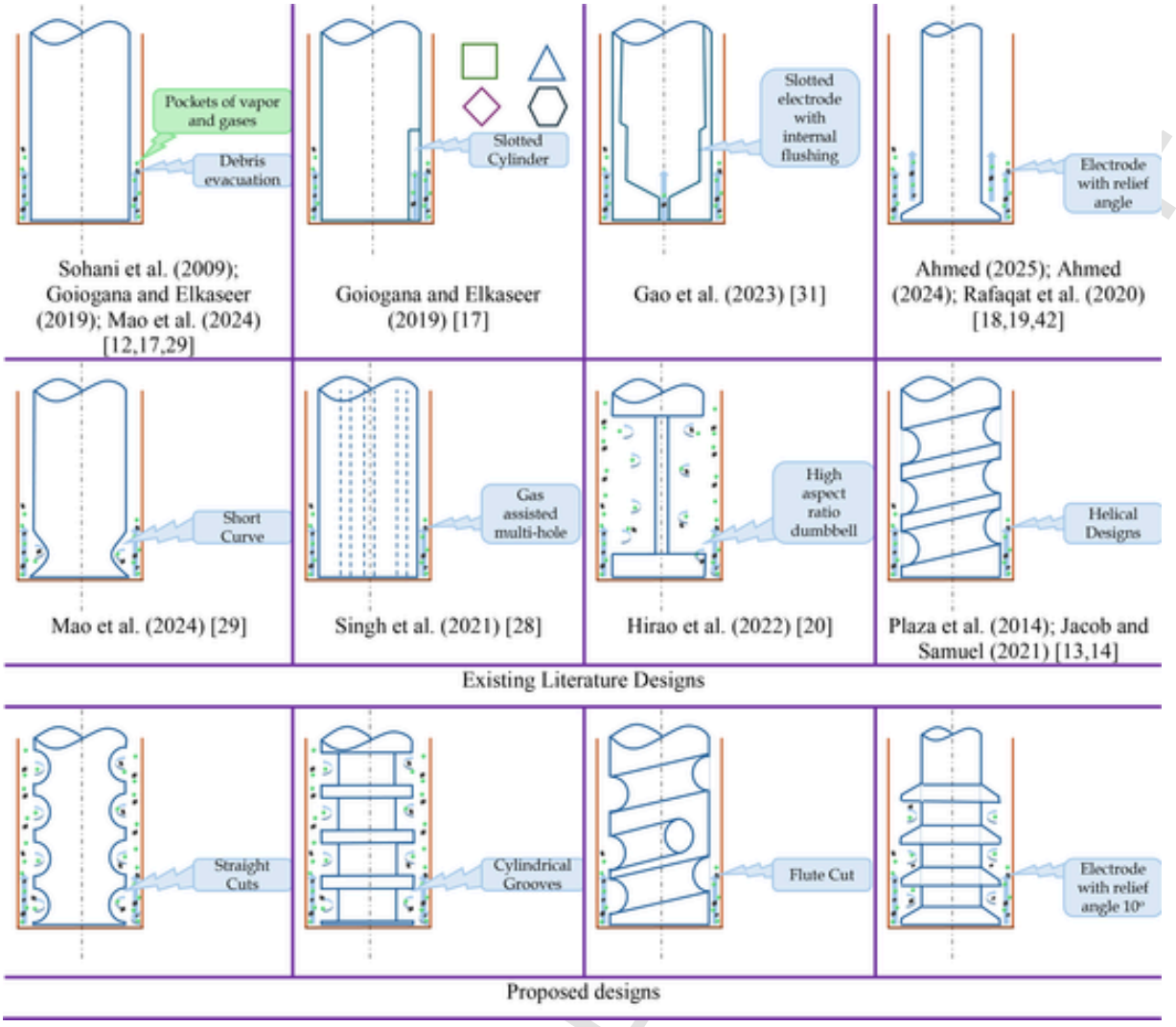

Fig. 1. Schematic illustrations of select electrode geometries investigated in die-sinking EDM literature and proposed in the present study.

gated influence of basic tool shapes i.e. triangular, square, rectangular, and circular cross sections in addition to variation of discharge current, pulse on-time, pulse off-time, and tool cross sectional area on MRR and tool wear rate (TWR) in die-sinking EDM. Circular cross-section geometry was found to yield the highest MRR and lowest TWR [12]. Modified tool geometries have been studied with numerous geometries including helical designs [13,14]. For blind hole drilling in Ti6Al4V with the simple single slotted electrode with eccentric micro-slot provided an increase in aspect ratio of 150% and providing straighter holes for pin like hole structures [15]. Novel shapes such as the U-shaped electrode and the self-sharpening electrode designs have also been studied [16]. Efficacy of self-flushing in EDM in terms of TWR, MRR and hole taper angle has also been studied. The largest hole taper angle was produced for the sharp-cornered triangular electrode (8.114°), and the smallest value (1.040°) was achieved with the cylindrical electrode. However, low hole taper angles (1.292°) were also obtained with the slotted cylindrical and round-cornered hexagonal electrodes. Furthermore, the investigation has shown that the round-cornered hexagonal electrode exhibited the maximum MRR (0.1257 g/min), while the cylindrical electrode gave the minimum MRR (0.0983 g/min) [17]. The effect of different tool designs for different materials for die-sinking EDM was studied with the tools made by pure copper (Cu), copper-tungsten (CuW), and graphite (Gr) were examined with varying relief angles. Machining time (MT), relative electrode wear (REW), diametric overcut (DOC), hole taper, and surface roughness with a percentage reduction of 47.28%, 91.29%, 37.35%, 9.4%, and 38.68%, respectively [18]. Effects of different electrode designs and materials for EDM of D2 steel have also been researched. This study's results indicate reduction in the machining time by 45% through modified tool designs. Likewise, Cu and CuW remarkably improve the MRR by 80% [19]. The effect of electrode shape on high aspect ratio deep hole drilling was also studied, indicating that the stirrer-shaped electrode was able to machine approximately 3.5 times deeper [20]. Researchers also demonstrated that a single notch micro electrode improved flushing by creating an unbalanced flow of dielectric medium that forced upstream flow. This design sink into deeper holes as compared to conventional cylindrical designs. Two-notch design on the other hand did not enhance flushing indicating asymmetric notch was the key to improving flushing and drilling deeper holes with EDM [21].

Beyond geometry, use of composite materials and advanced manufacturing techniques for the development of electrodes is another interesting avenue [22]. Using materials such as ZrB_2-Cu electrode (ZrB_2 40 wt% Cu) is another non-conventional electrode tool design strategy produced from powder metallurgy composite. This powder metallurgy electrode enhanced MRR and reduced tool wear rate when compared to pure copper electrode on EDM of mild steel. A hard ZrC layer was formed on composite electrode that enhanced wear resistance of electrode. This approach led to higher surface roughness and diametrical

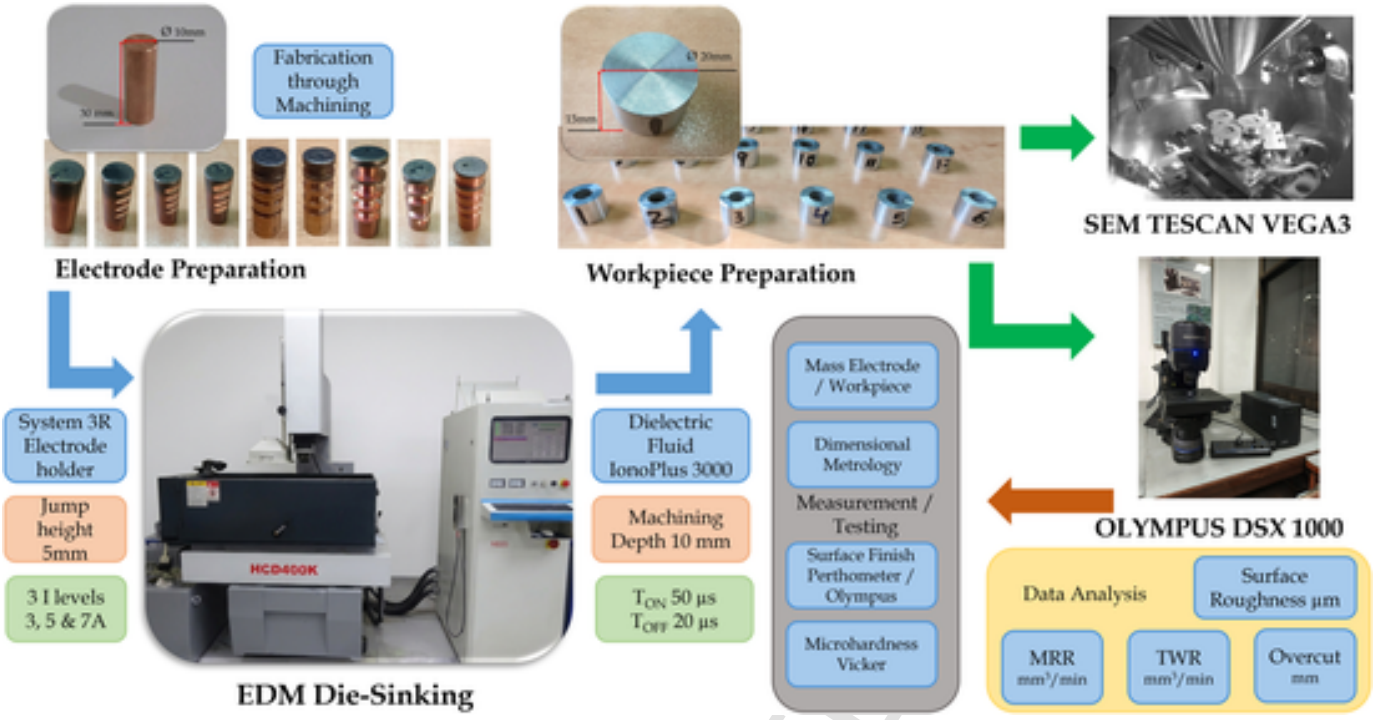

Fig. 2. Schematic of the experimental procedure.

Table 1
Chemical composition of Al2219 (wt.%).

Alloy	Al	Cu	Mn	Zr	Fe
Al2219	92.25-93.90	5.8-6.8	0.2-0.4	0.1-0.25	≤0.3

Table 2
Density computations of Al2219 cylindrical electrode samples from direct mass and geometry measurements.

Sample No.	Diameter (mm)	Length (mm)	Mass (g)	Volume (cm^3)	Density (g/cm^3)
Sample 09	20.119	15.040	13.557	4.781	2.835
Sample 12	20.063	15.015	13.458	4.747	2.835
Sample 14	20.129	14.923	13.463	4.749	2.835
Sample 18	20.538	15.040	14.126	4.983	2.835

Table 3
Properties of electrode and workpiece.

Properties of Material	Copper	Al2219
Density (g/cm^3)	8.96	2.84
Melting Temperature (°C)	1085	543-643
Thermal Conductivity (W/mK)	385	120-130
Electrical Conductivity (IACS%)	100	30-31
Resistivity (Ω.m)	$1.72x10^{-8}$	$5.7x10^{-8}$
Hardness (HRC)	33	72 HRB

overcut [23]. Electrodes made from two non-convention rapid tooling approaches have also been investigated, out of which one approach utilized fused filament fabrication in which copper was deposited by electroplating on this 3D printed electrode [24]. The other approach was fabrication of tool electrode from casting of copper. Electrodes fabricated through fused filament fabrication on polycarbonate cores performed almost similarly to solid copper cast electrodes. In addition, fused deposition modeling (FDM) 3D printed with acrylonitrile butadiene styrene (ABS) material has also been studied [25]. These 3D printed ABS electrodes were used in electrical discharge machining of mild steel after copper coating from electrolysis process, evaluating MRR and SR using Taguchi L25 design and ANOVA. Electrode production in EDM accounts for 50% of operational cost and time during whole process. Therefore, fabrication of tool electrode from FDM and subsequently metallization of FDM fabricated electrode and controlling the coating thickness can reduce the cost and time. In terms of MRR, TWR and SR these tool electrodes have same performance as solid electrodes produced conventionally [26].

Recent research has explored hybrid and assisted EDM approaches to overcome inherent process limitations. Cakiroglu performed EDM turning on a Die Sinking Machine using a motorized pulley mechanism to rotate the workpiece while electrode profile is sunk into the rotating workpiece. This modified EDM and electrode setup allowed for the turning operation in an EDM machine [27]. Singh [28] worked for improved flushing through non-conventional electrode design. Three different approaches were studied, conventional stationary tool, solid rotary tool and third one with gas assisted multi-hole rotary tool electrode. Compressed argon gas was used. 70% improvement in MRR was observed in multi hole gas assisted approach when compared with conventional electrode and 30% TWR was improved. For rotation of electrode a pulley-motor system was used instead of in-built c-axis of the machine. Various other parameters such as gas pressure, electrode speed, current, pulse on time and pulse off time were studied. Using stepped electrodes in conjunction with injection flushing has also been utilized. The MRR was increased by 75% and 82% when using cylindrical electrodes with pressures of 120 psi and 40 psi, respectively [29]. In contrast, the MRR with injection flushing was about 80% of that without injection flushing when using stepped electrodes. Furthermore, the feasibility of the novel Maglev EDM proved suitable with an average MRR of 40 µg/min [30]. Another, concept of note is the use of internal flushing which has reported showing with the increase in pulse width, the MRR increases first and then decreases, while the TWR decreases gradually [31]. The study of channel and gap flow is another niche that has been explored along with electrode jump function, showing a significant impact on the liquid flow of cavitation, as well as the presence of vortex gaps in some cases [32,33].

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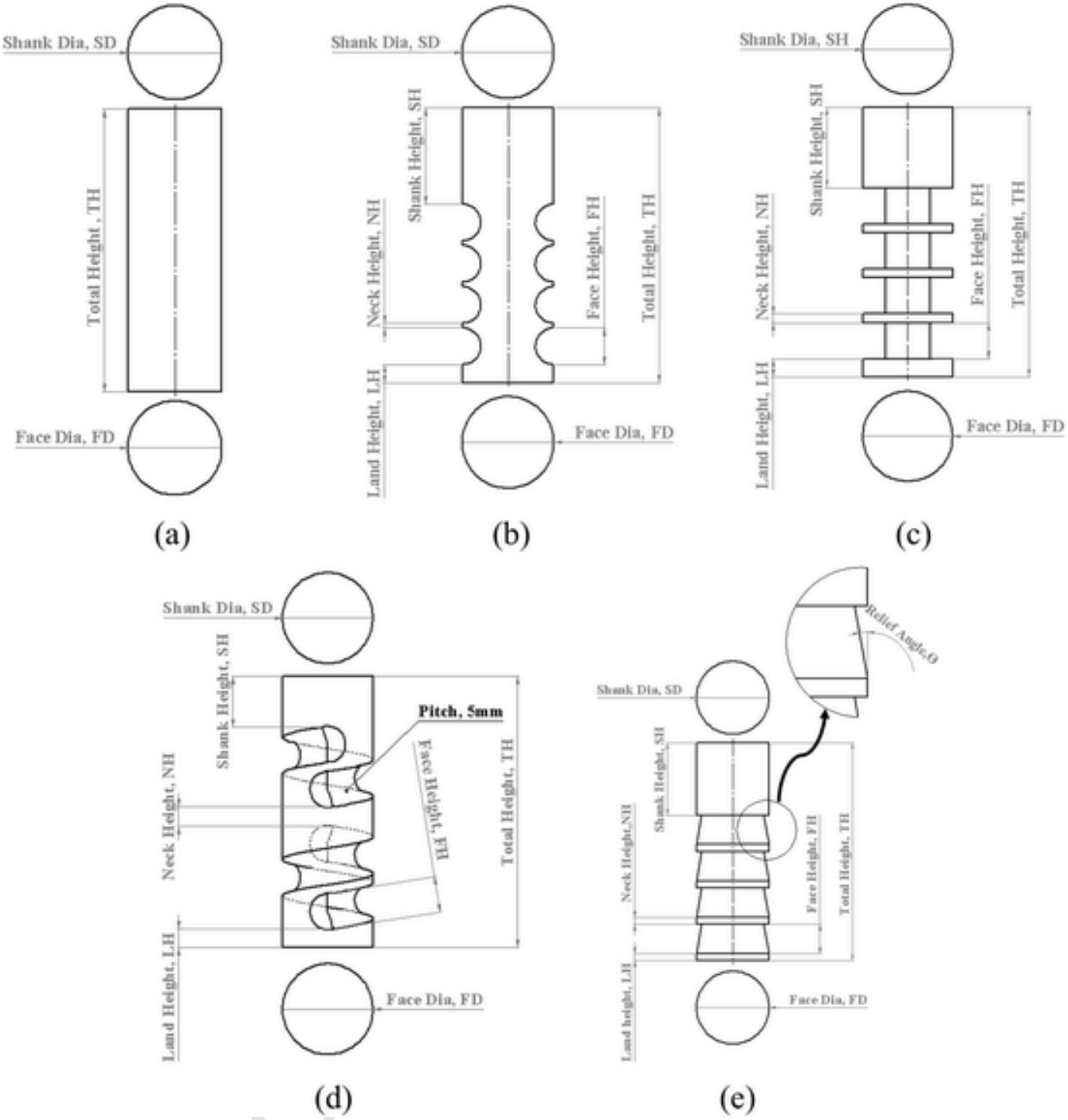

Fig. 3. Schematic diagram of Electrodes (a) Conventional Design (b) Straight radius cut designs (c) Groove shaped designs (d) Flute shaped design (e) Relief Angle 10° design.

Other directions while using EDM for this include use of AI for enhancing machining performance [34], hybrid systems with ultrasonic [35] or magnetic field assistance [36], and the utilization of novel fluids with sustainability for better environmental preservation [37,38]. Plasma flushing efficiency (PFE) on heat affected zones (HAZ) and solidified white layer thickness (WT) during micro EDM is another interesting direction [39]. The material used in the particular study was Inconel-718. The results reveal that extending the pulse-on duration increases WT and HAZ depth owing to longer temperature exposure during discharge. Furthermore, numerical modeling of MRR through actual discharge power during EDM operation provided interesting perspective [40]. SEM and EDS analysis indicated the presence of different anomalies and confirmed surface alloying. A maximum deviation of 13.05% from practical evidence was observed using their model. Researchers have also presented a method for removing EDM oxides and damage [41].

As discussed, die sinking EDM has undergone significant development in process parameters, surface integrity and MRR improvements. However, a systematic assessment of flushing-driven electrode geometries [42], specifically radial cuts of varying diameters, grooves, flutes,

Table 4
Dimensional characteristics of non-conventional tool electrodes, DNC.

Design	LH (mm)	FH (mm)	NH (mm)	SH (mm)	Design mass (g)
D_{NC-1}	3	2	2.5	11.5	20.55 (97.99% of D_c)
D_{NC-2}	2.5	3	1.5	11	19.83 (94.56% of D_c)
D_{NC-3}	2	4	0.5	10.5	18.66 (88.98% of D_c)
D_{NC-4}	3	2	3	10	16.78 (81.65% of D_c)
D_{NC-5}	2.5	3	2	9.5	14.68 (71.43% of D_c)
D_{NC-6}	2	4	1	9	12.58 (59.99% of D_c)
D_{NC-7}	2	4	2	5.6	17.05 (81.30% of D_c)
D_{NC-8}	1	4	1	10	19.47 (92.84% of D_c)

Table 5
EDM pulses run on each electrode.

S. No.	Current I(A)	Gap Voltage V (V)	Pulse on time T_{on} (μs)	Pulse off time T_{off} (μs)	Retract Height (mm)
1	3	80	50	20	5
2	5	80	50	20	5
3	7	80	50	20	5

and relief angles, and their relationship to MRR, TWR, dimensional overcut, and surface integrity remains underexplored, particularly for aluminium alloy Al2219 in blind hole configurations. The present work addresses this gap by designing and fabricating nine distinct electrode configurations, including a conventional cylindrical reference and eight modified geometries featuring repeated pattern segments. This investigation evaluates how geometric modification influences flushing efficiency, and how these factors translate into measurable differences in machining performance and surface quality. Surface integrity is assessed using both two-dimensional (Ra) and three-dimensional (Sa) roughness parameters at the bottom of blind holes, a region traditionally inaccessible to stylus-based instruments and therefore often overlooked in EDM literature. A radar plot-based multi-performance analysis is further employed to identify electrode designs that offer the best balance across competing performance metrics under different current conditions. Fig. 1 provides a schematic overview of the electrode geometries reported in the literature alongside the nine novel configurations designed and fabricated for the present investigation. This work systematically investigates and characterizes the influence of varied electrode designs on machining performance and surface integrity dur-

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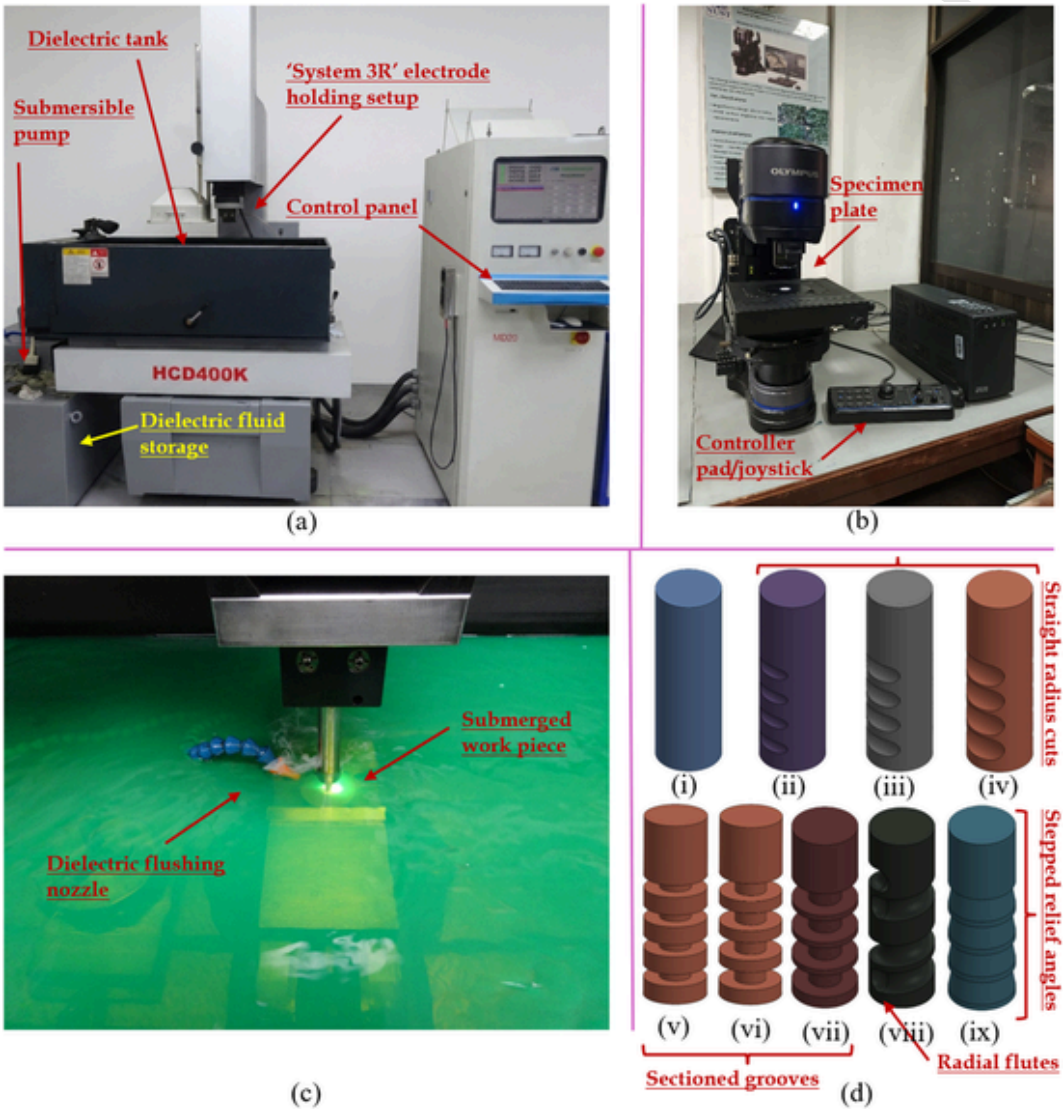

Fig. 4. Experimental equipment and tooling used in this study: (a) Die-Sinking EDM used in experiments. (b) Olympus DSX1000 for sample scanning. (c) Experimental setup with workpiece submerged in dielectric fluid. (d) Set of custom-fabricated copper electrodes: (i) Conventional cylinder (ii) Straight cut φ2mm (iii) Straight cut φ3mm (iv) Straight cut φ4mm (v) Groove 2 mm wide (vi) Groove 3 mm wide (vii) Groove 4 mm wide (viii) Flute 4 mm width (ix) 10° Taper/relief angles.

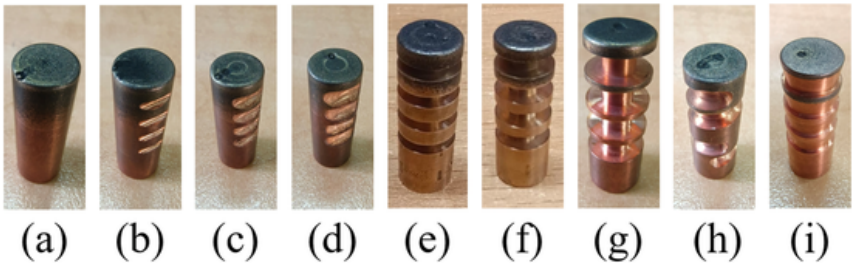

Fig. 5. Used Electrodes in Die-Sinking EDM Experiments (a) Conventional Design (b) Dia 2 mm Design (c) Dia 3 mm Design (d) Dia 4 mm Design (e) Groove 2 mm wide design (f) Groove 3 mm wide design (g) Groove 4 mm wide design (h) Flute 4 mm wide Design (i) 10° Relief Angle Design.

Table 6
Total number of experiments with input parameter details.

Exp. No.	Pattern name (Electrode)	I (A)	V (V)	T_{on} (μs)	T_{off} (μs)	Retract Height (mm)	Electrode shape
1	Conventional	3	80	50	20	5	
2	D_C	5	80	50	20	5	
3		7	80	50	20	5	
4	Straight cut	3	80	50	20	5	
5	φ2 mm	5	80	50	20	5	
6	D_{NC-1}	7	80	50	20	5	
7	Straight cut	3	80	50	20	5	
8	φ3 mm D_{NC-2}	5	80	50	20	5	
9		7	80	50	20	5	
10	Straight cut	3	80	50	20	5	
11	φ4 mm D_{NC-3}	5	80	50	20	5	
12		7	80	50	20	5	
13	Groove 2 mm	3	80	50	20	5	
14	wide	5	80	50	20	5	
15	D_{NC-4}	7	80	50	20	5	
16	Groove 3 mm	3	80	50	20	5	
17	wide	5	80	50	20	5	
18	D_{NC-5}	7	80	50	20	5	
19	Groove 4 mm	3	80	50	20	5	
20	wide D_{NC-6}	5	80	50	20	5	
21		7	80	50	20	5	
22	Flute 4 mm	3	80	50	20	5	
23	wide	5	80	50	20	5	
24	D_{NC-7}	7	80	50	20	5	
25	10° Relief	3	80	50	20	5	
26	Angle	5	80	50	20	5	
27	D_{NC-8}	7	80	50	20	5	

ing die-sinking electrical discharge machining of blind holes in Al2219. Through this systematic approach, the study establishes relationships between electrode geometry, process performance parameters, and the resulting surface characteristics, offering practical guidelines for electrode selection in industrial blind hole machining applications.

2. Methodology

Tool electrodes of different geometric designs were employed and their effects on major performance parameters such as machining time, TWR, MRR, surface roughness (SR) and dimensional overcut were investigated (Fig. 2). EDM Die-Sinking was carried out on Al2219 alloy. Al2219 represents high specific strength and heat resistance, that is why it finds its applications in automobile and aerospace industry. T6 thermal heat condition of Al2219 enhances its ductility, hardness and strength due to its unique precipitation hardened microstructure [43] (Table 1). This Aluminum copper alloy is widely used in aerospace and automotive industry where it finds its application in fuel tanks & structural components due to its substantially high strength to weight ratio and its hardness upon heat treatment and subsequent artificial aging [44].

2.1. Samples preparation

In the present experimental setup, a cylindrical rod of Al2219 with diameter ≈20 mm was selected. This rod was then parted with the help of conventional lathe machine into 27 test pieces. A parting tool 2 mm wide was employed. All test pieces were then engraved with serial numbers with the help of scriber. Each prepared workpiece from lathe was employed for one complete experimental run. All samples were weighed before experimental run with OHAUS PR124 ZH/E Analytical Balance with a least count of 0.1 mg to accurately determine quantity of material removed. Properties of 4 random samples before experimentation are presented in Table 2. With mass and volume data, density of material was computed and found to be consistent for all samples.

2.2. EDM electrode design

Copper is used as an electrode in these experiments as it is the most widely used electrode material for die-sinking EDM due to its superior conductivity and machinability [45]. Properties of the electrode material [18] and Al2219 are presented in Table 3. Furthermore, 10-15 mm blind holes were machined using EDM. Therefore, electrodes length was set to 30 mm to conveniently hold electrodes in electrode holder.

Nine different electrode designs were manufactured in-house and applied to workpiece specimens including the conventional cylindrical design as a reference. All designs had repeated patterns up to the holding area and neck regions with no patterns in between as presented in Fig. 2. Spaces between repeatable patterns allow the electrode to be reusable after the face of electrode wears down. The patterns were intended to allow EDM to machine deeper holes with better MRR. This concept was also tested, and the results are discussed in later sections. Following tool electrode designs were selected for this study:

1 Conventional cylinder, D_C
2 Straight cut φ2mm, D_{NC-1}
3 Straight cut φ3mm, D_{NC-2}
4 Straight cut φ4mm, D_{NC-3}
5 Groove 2 mm width, D_{NC-4}
6 Groove 3 mm width, D_{NC-5}
7 Groove 4 mm width, D_{NC-6}
8 Flute 4 mm width, D_{NC-7}
9 10° Taper, D_{NC-8}

The cavity patterns on these electrodes were developed using milling cutters of diameter 2 mm, 3 mm & 4 mm. Grooved electrode $D_{NC-(4-6)}$ was manufactured using grooving tool of 1.5 mm. Flute pattern was machined on electrode using ball nose milling cutter of radius 2 mm using 5-axis milling machine.

With reference to design strategy, electrodes used can be grouped in four different shapes namely Radial cut, Groove shaped cut, Flute shaped cut, and Relief angle (10°). Depth of 10 mm and blind holes were selected as different electrode designs have not been studied for deep machining where effects of flushing with dielectric fluid becomes

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Table 7
Samples of Al2219 machined with different electrode designs.

Design Of Electrode	Machined Hole Samples		
	Pulse 1	Pulse 2	Pulse 3
Conventional			
Dia 2 mm cuts			
Dia 3 mm cuts			
Dia 4 mm cuts			
2 mm wide Groove			
3 mm wide Groove			
4 mm wide Groove			
4 mm wide Flute			
10° Relief Angle			

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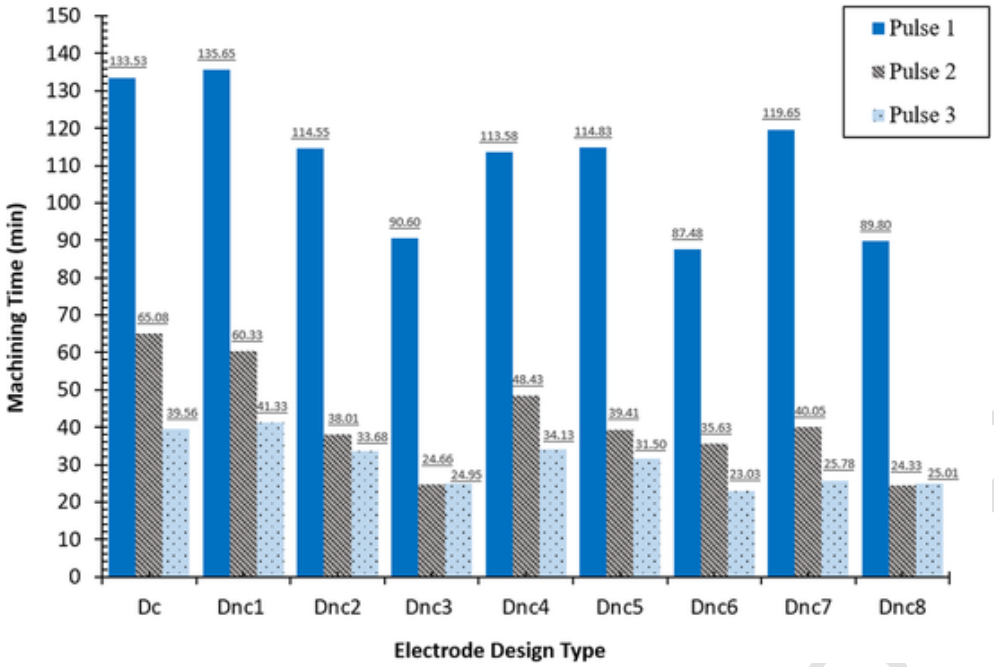

Fig. 6. Comparison of Machining time for different Tool Electrodes Dc and DNC.

Table 8
Experimental results of this study.

Experiment No.	Pattern on Electrode	Machining Time (min)	MRR (mm^3/min)	TWR (mm^3/min)	Overcut (mm)	Surface Roughness		
						Cylindrical Surface Ra (μm)	Bottom Surface Ra (μm)	Bottom Surface Sa (μm)
1	Conventional	133.53	6.255	0.013	0.13	2.436	1.764	2.987
2	Dc	65.08	13.035	0.049	0.17	3.192	5.570	5.856
3		39.56	21.698	0.139	0.19	3.265	3.132	2.711
4	Dia 2 mm cuts	135.65	6.189	0.012	0.13	2.626	1.976	3.040
5	Dnc-1	60.33	14.145	0.068	0.17	3.169	2.550	5.255
6		41.33	20.773	0.147	0.21	3.450	7.788	9.371
7	Dia 3 mm cuts	114.55	7.320	0.014	0.15	2.268	2.402	3.083
8	Dnc-2	38.01	17.956	0.084	0.20	2.064	2.602	4.087
9		33.68	25.499	0.196	0.22	3.968	6.141	6.529
10	Dia 4 mm cuts	90.6	9.172	0.019	0.13	2.245	1.693	2.791
11	Dnc-3	24.66	33.920	0.162	0.16	3.112	5.263	5.675
12		24.95	33.828	0.260	0.19	4.544	4.297	3.564
13	2 mm wide Groove	113.58	7.333	0.028	0.15	3.076	1.209	2.294
14	Dnc-4	48.43	17.169	0.153	0.19	3.285	2.232	4.392
15		34.13	24.619	0.268	0.22	3.993	6.202	9.112
16	3 mm wide Groove	114.83	7.243	0.042	0.13	3.120	0.492	4.233
17	Dnc-5	39.41	21.349	0.201	0.17	3.463	6.633	6.239
18		31.50	26.541	0.331	0.19	4.172	4.608	6.841
19	4 mm wide Groove	87.48	9.579	0.023	0.14	3.154	1.897	2.464
20	Dnc-6	35.63	23.741	0.109	0.17	3.636	3.907	8.005
21		23.03	36.742	0.257	0.18	4.476	2.942	8.516
22	4 mm wide Flute	119.65	6.922	0.013	0.12	2.390	1.576	1.759
23	Dnc-7	40.05	21.065	0.099	0.17	3.107	5.894	6.945
24		25.78	33.076	0.252	0.20	4.001	4.302	5.803
25	10° Relief Angle	89.8	9.266	0.018	0.13	2.561	1.204	2.342
26	Dnc-8	24.33	34.496	0.125	0.16	3.521	5.218	5.478
27		25.01	33.730	0.242	0.17	4.342	6.446	6.384

very limited. Internal features in Al2219 components such as deep guide holes, precision passages, precise guideways for fasteners, alignment pins, fluid routing and precise sensor integration spaces, holes often extend to 8-12 mm or more in depth with tight dimensional tolerance and surface finish requirement. The 1st basic design is taken as a reference. Furthermore, all electrodes were fabricated from same rod of copper and using conventional machining operations such as cutting, facing, turning, parting, grooving and milling. Conventional milling was required only for one non-conventional flute shaped electrode design. All other non-conventional designs required only lathe and drilling machine operations.

Non-conventional electrodes are designed with repeated patterns along the length of the electrodes as we move from Landing segment towards the shank of the tool electrode. Equation (1) was developed for this study to define geometric architecture of proposed electrode designs:

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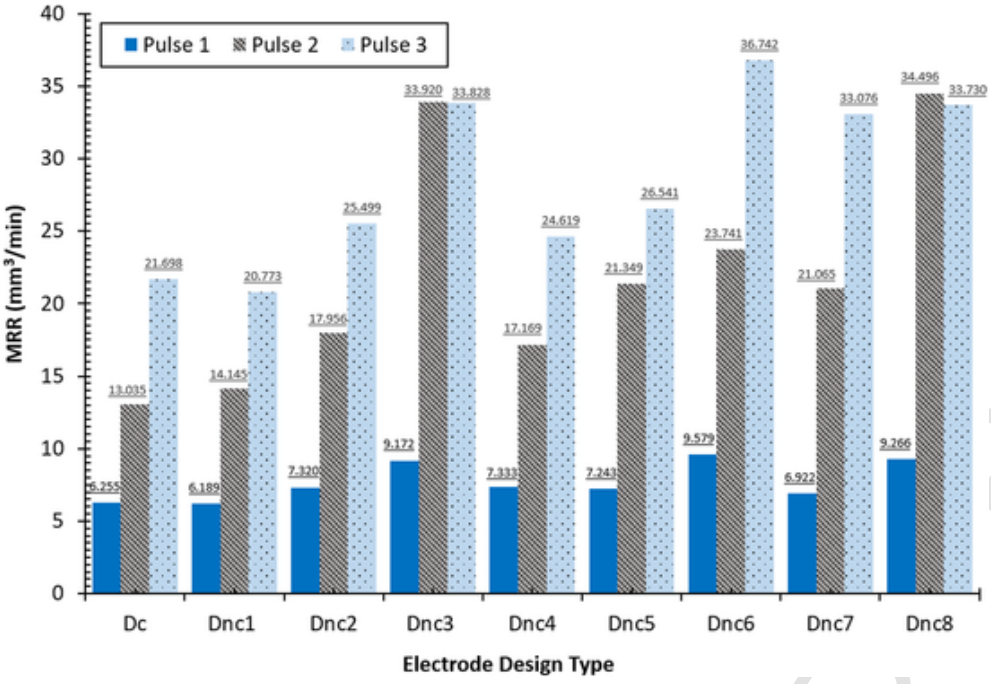

Fig. 7. Comparison of MRR for different tool electrode designs.

Table 9
Replicated MRR values (Mean ± SD) and 95% CI.

Design	I (A)	Mean ± SD (mm^3/min)	95% CI
Dc	3	6.238 ± 0.022	6.185-6.292
Dc	5	13.148 ± 0.127	12.831-13.464
Dc	7	21.603 ± 0.085	21.392-21.814
DNC-6	3	9.567 ± 0.045	9.455-9.678
DNC-6	5	23.599 ± 0.147	23.233-23.964
DNC-6	7	36.431 ± 0.292	35.706-37.155
DNC-8	5	34.712 ± 0.204	34.207-35.218

Table 10
One-way ANOVA at 5A (Dc vs DNC-6 vs DNC-8).

Source	SS	Df	MS	F	p	η^2
Between groups	697.772	2	348.886	13192.2	< 0.0001	0.9998
Within groups	000.159	6	0.02645	-	-	-
Total	697.931	8	-	-	-	-

Table 11
Tukey HSD pairwise comparisons at 5 A.

Comparison	Mean difference (95% CI)	% improvement (95% CI)	Adjusted p
DNC-6 vs Dc	10.451 (10.044-10.858)	79.5% (76.4-82.6%)	< 0.001
DNC-8 vs Dc	21.565 (21.157-21.972)	164.0% (160.9-167.1%)	< 0.001
DNC-8 vs DNC-6	11.114 (10.706-11.521)	47.1% (45.4-48.8%)	< 0.001

$$Total\ Height\ (TH) = LH + \sum FH + \sum NH + SH \quad (1)$$

Where, LH is the height of landing segment. FH is the height of face, NH is the height of Neck, and SH is the height of shank. AH, NH and SH in the said designs were varied while total height of the tools was kept constant at 30 mm. NH regions are employed repeatedly across all electrodes in this work because they enhance electrode reusability when machining multiple holes in sequence, a common operation in automotive and aerospace industries [46]. In an industrial setting, cost efficiency is a primary driver which, when coupled with rapid production, yields significant dividends. These repeatable patterns on the electrodes act as future face diameter areas, after the electrode wears down from the tip. Once wear occurs, the electrode can be faced with the help of lathe down to the NH area, turning that region into a new face diameter. This practice potentially improves the overall reusability of electrode manifolds. This uniqueness therefore offers efficiency in terms of both time and cost. Flute shaped design has a pitch of 5 mm. Electrode with relief angle design also has similar strategy of repeated NH regions to improve reusability. The remaining parameters such as LH, FH, NH, SH are kept similar within tolerances across all the designs. A detailed representation of the electrode design is presented in Fig. 3.

Design mass of conventional tool electrode, D_c is 20.97 g and its total height is 30 mm and has constant 10 mm diameter. Design mass of all other non-conventional electrodes, D_{NC} in comparison with D_c is given in Table 4 in the form of percentage of D_c. Total height (TH) of all electrodes is kept constant at 30 mm in this study. Shank diameter (SD) and face diameter (FD) are 10 mm for all conventional and non-conventional electrodes.

2.3. Experimental setup

For this study, die sinking EDM of Hanchuan Machine tool HCD-400 is used and synthetic dielectric fluid ionoplus-3000 was deployed in EDM tank for flushing due to its more occupational health and safety friendly composition as compared to kerosene which is a more common counterpart. Each of nine different electrodes are applied with three different current levels 3A, 5A & 7A. Other machining parameters such as Gap Voltage (V), Pulse time-on (T_{on}, μs), Pulse time-off (T_{off}, μs) & Retract Height (mm) were maintained constant during these experiments. Table 5 presents the corresponding pulses that were run on each electrode design to study the behavior of corresponding patterns on these electrodes.

The electrode was connected as cathode and the workpiece as anode, which is the standard polarity for die-sinking EDM. Dielectric fluid (Ionoplus-3000) was supplied through external flushing nozzles at a constant flow rate of 10.34 L/min, and the dielectric temperature was maintained at 20 ± 1 °C. The servo-controlled electrode jump was set to 5 mm height and activated every 3 s to help debris evacuation, with servo reference voltage and gain kept constant across all experiments.

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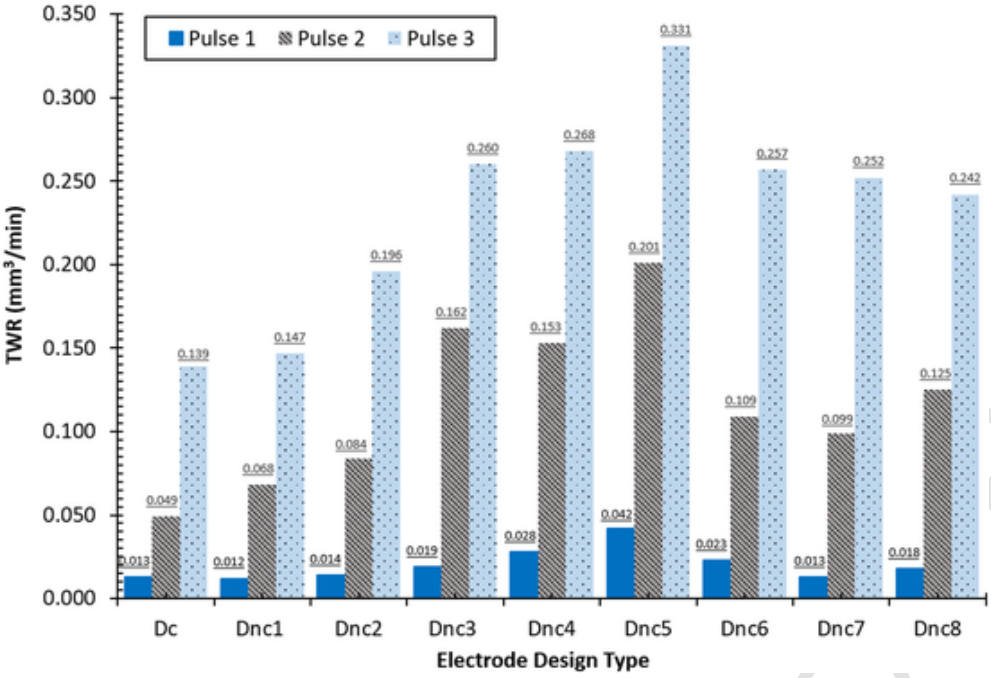

Fig. 8. Comparison of TWR for different tool electrode designs.

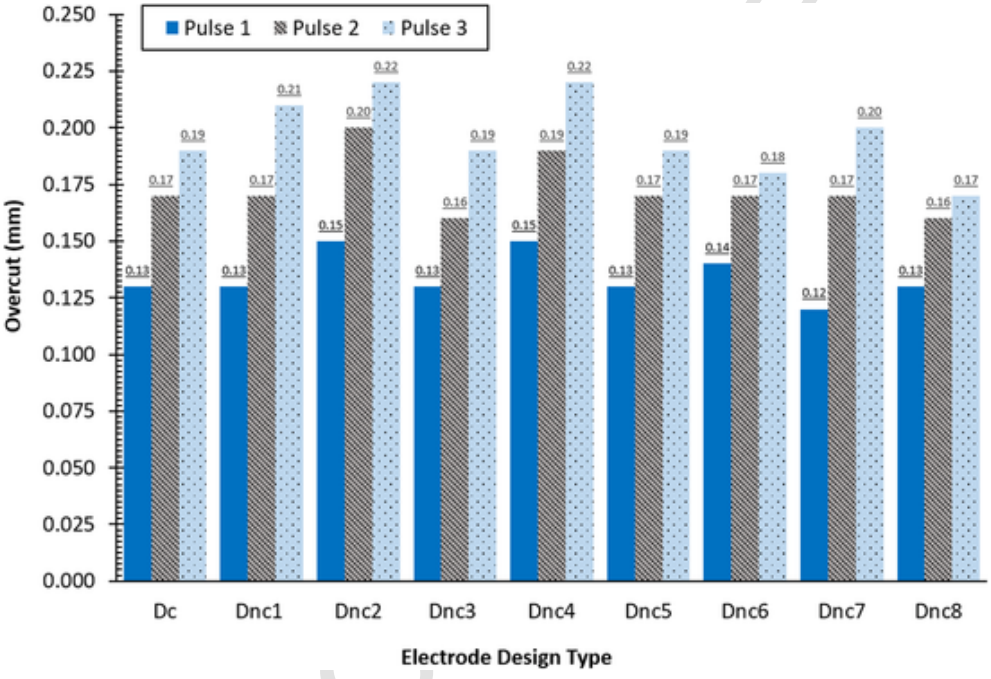

Fig. 9. Comparison of overcut for different tool electrode designs.

Machining depth was set to 10 mm using the machine Z-axis control, and depth was verified after machining with a vernier caliper, with deviation 09.94-10.07 mm for all runs. Electrode alignment was ensured using System 3R work holding system with locating pins, giving repeatable X, Y and Z positioning and eliminating manual referencing errors.

In each experiment, both the electrode and workpiece were weighed before as well as after machining. Furthermore, size of electrode was measured before each experiment and length reduced during the machining was measured. Machining time, MRR, TWR, SR and dimensional overcut were the performance parameters examined for all different electrode tool designs. Mass difference of the electrode, before and after the machining along with material density data were used to calculate TWR of each experiment. Size reduction data alone was insufficient to calculate non-uniform tool wear at machining face of electrode due to micro-pitting difficult to measure conventionally. TWR was calculated by applying following equation [47]:

$$TWR = \frac{m_{TB} - m_{TA}/\rho_{Cu}}{t_m} \quad (2)$$

Where,

m_{TB} is tool mass before
m_{TA} is tool mass after
ρ_{Cu} is density of copper

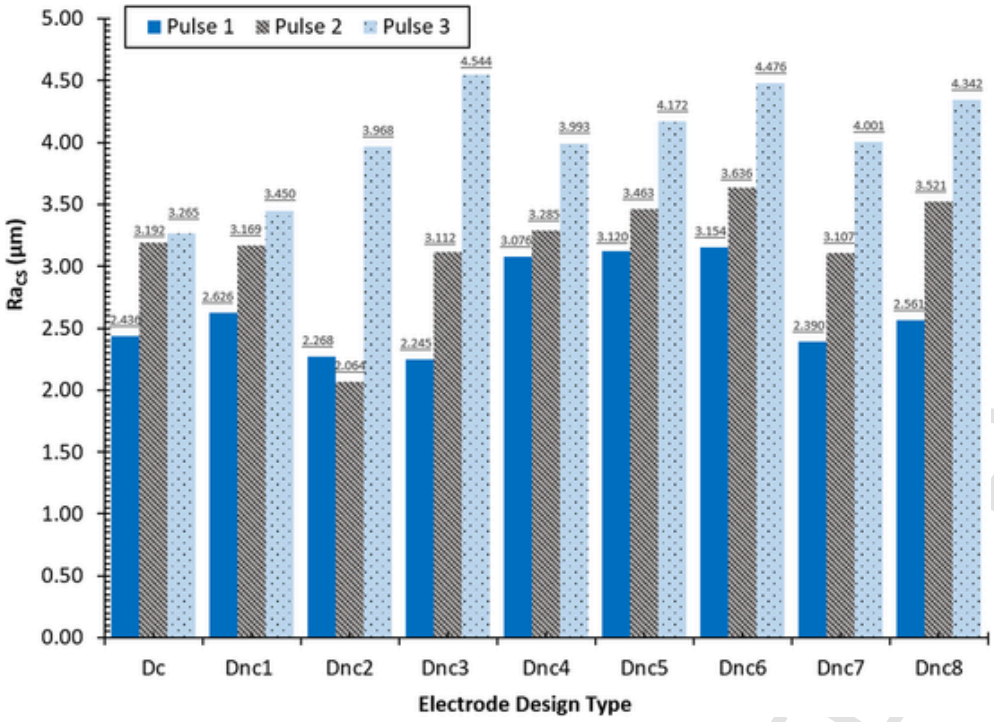

Fig. 10. Comparison of Cylindrical Surface Ra for different tool electrode designs.

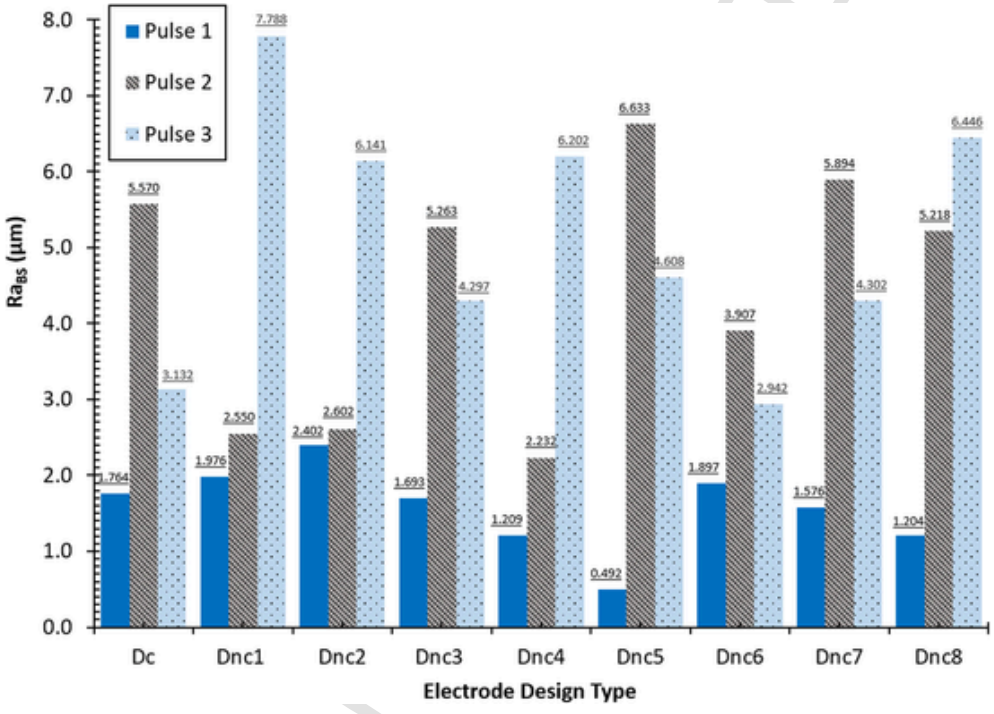

Fig. 11. Comparison of bottom surface Ra of electrode designs.

t_m is the machining time.

Mass difference of the workpiece samples, before and after were used to calculate MRR to cater for the fact that at the base of machined holes there can be uneven surface. MRR was calculated by applying following equation [34]:

$$MRR = \frac{m_{WB} - m_{WA} / \rho_{Al2219}}{t_m} \quad (3)$$

Where,

m_{WB} is tool mass before

m_{WA} is tool mass after

ρ_{Al2219} is density of Al2219.

The overcut of all experimental runs was calculated by the radial difference between the machined hole diameter and the electrode diameter. The equation is [48]:

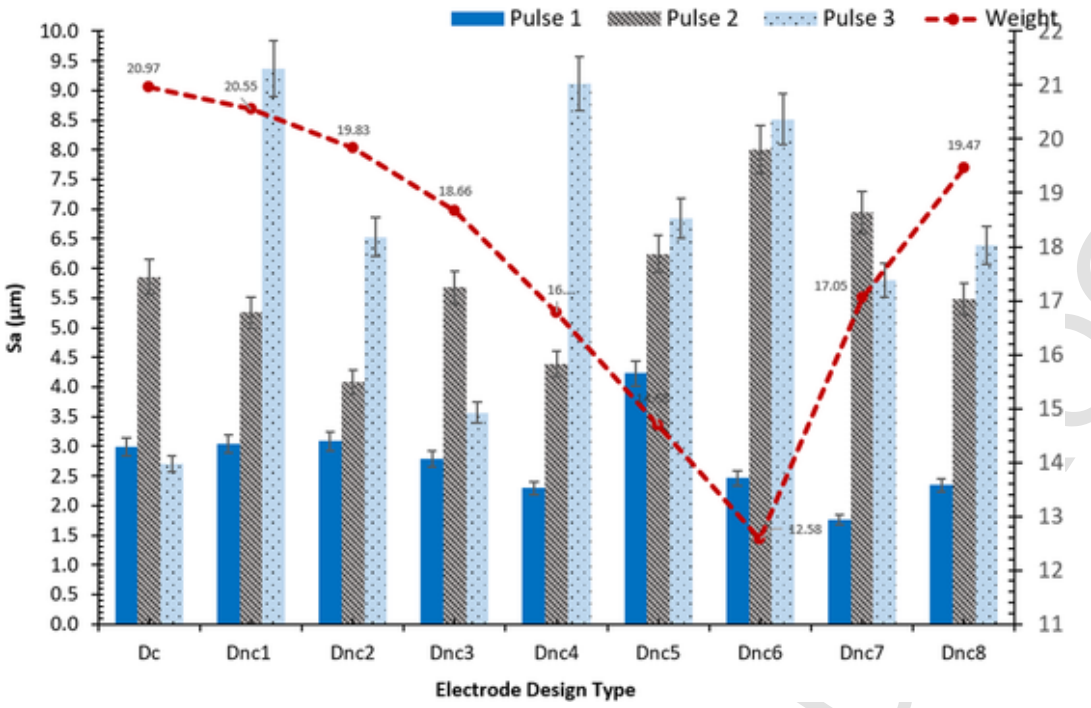

Fig. 12. Comparison of Sa along with design mass of Electrode Designs.

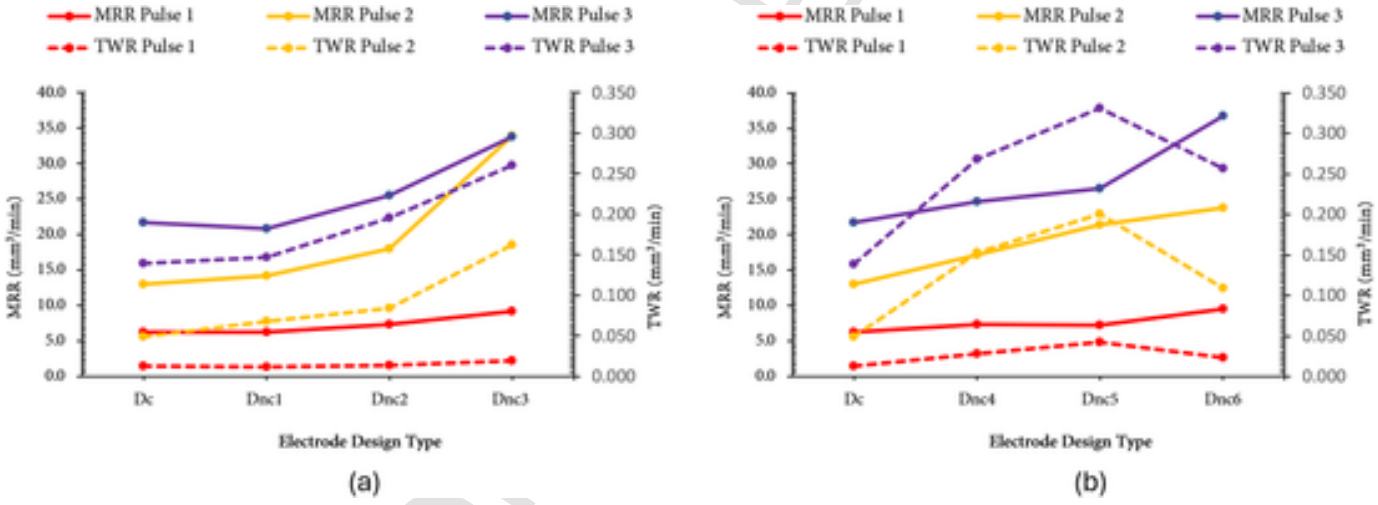

Fig. 13. Relative Comparison of TWR and MRR for (a) Diameter Cuts Electrode (b) Groove cuts Electrode.

$$O_c = \frac{D_{mh} - D_e}{2} \quad (4)$$

Where,

O_c is the overcut.

D_{mh} is the machined hole diameter.

D_e is the electrode diameter.

The die-sinking EDM setup used in this work incorporated a System 3R work holding system, which provided a precise, repeatable, and efficient interface for electrode mounting. System 3R modular reference system such as locating pins, enabled consistent X, Y, and Z positioning of each electrode relative to the machine's coordinate axes. By employing a collet-based holder assembly, the system facilitated rapid electrode changes, ensuring that each electrode maintained positional repeatability across multiple machining trials (Fig. 4a). The setup on the right side in Fig. 4a shows machine's control panel and the left side of the figure shows storage of dielectric fluid with submersible pump. This setup eliminated the need for repeated manual referencing and significantly enhanced the overall repeatability and dimensional accuracy of the deep blind holes machined in Al2219. This arrangement helped maintain consistent hole geometry and improved the overall reliability of the experimental results.

The evaluation of surface finish at the bottom of a blind hole with depth greater than 2-3 mm presents a distinct metrological challenge, as conventional stylus-based profilometers cannot access the confined geometry without risking damage to the measurement probe or cutting the side walls. To overcome this limitation, the present study employed an Olympus DSX1000 digital microscope for non-contact surface roughness measurement at the base of each machined blind hole comprising 10 mm diameter and 10 mm depth. The Olympus DSX1000 offers a magnification range of 20x to 7,000x and is equipped with advanced 2D and 3D surface profiling capabilities, including height measurement

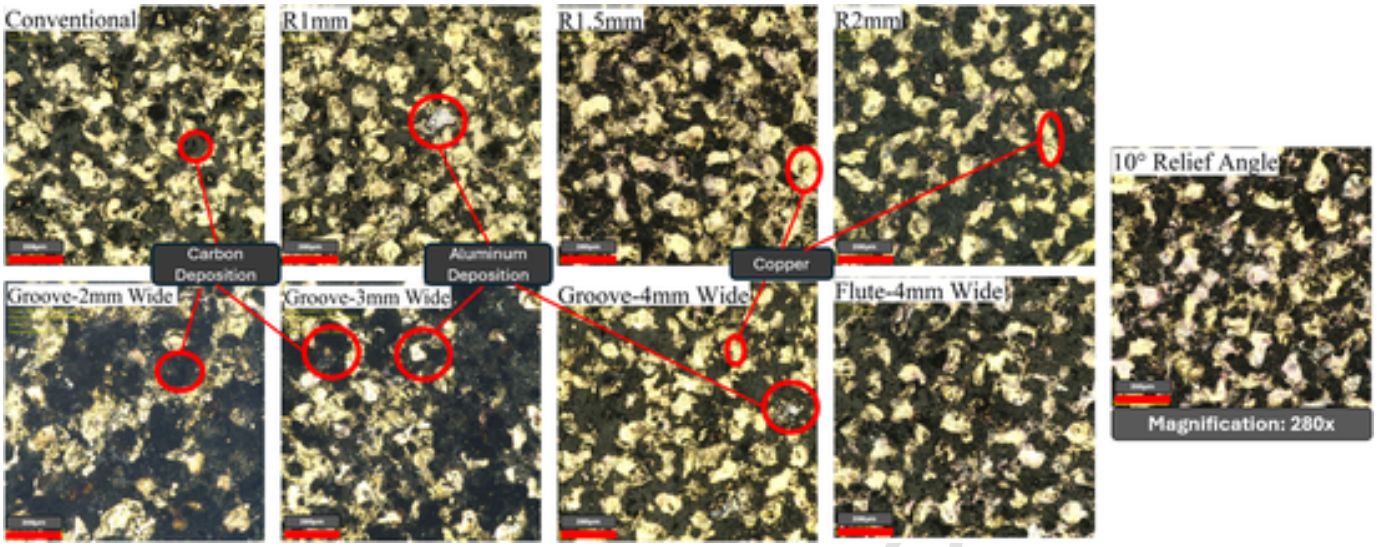

Fig. 14. Microstructural analysis of worn electrodes.

and roughness analysis through optical sectioning and focus variation (Fig. 4b).

For each specimen, the images were acquired at different focal planes to reconstruct a three-dimensional topographic map. From this reconstructed surface, standard roughness parameters, namely arithmetic mean line roughness (Ra) and three-dimensional areal roughness (Sa), were computed using the instrument's integrated software. While Ra provides a conventional line-based assessment of surface texture, Sa is necessary to capture the full areal topography; including peaks, valleys, and spatial variations across the entire bottom of hole. This study was conducted using 9 different tool electrodes under 3 different pulses (Fig. 5). In each pulse, the current was varied while other parameters were kept constant as MRR majorly depends on current (Table 6).

R_a of the cylindrical wall was measured with Mahr M2 perthometer. Probe of perthometer could easily reach inside diameter 10 mm hole so requirement to cut open workpieces was eliminated. The two-dimensional R_a along the cylindrical walls of the machined blind holes was measured. Unlike the bottom surface which required an optical method, the vertical cylindrical wall of the 10 mm diameter hole provided sufficient access for the instrument's stylus probe. The machined samples are presented in Table 7.

3. Results

3.1. Machining time

The machining time for 9 electrodes under 3 pulse conditions was measured precisely using in-built mechanisms of the equipment (Table 8, Fig. 6). The shortest machining time, that is 23.03 min, was achieved with D_{NC-6} (4 mm Groove Design) running on pulse 3. This amounts to 41.78% reduction in machining time under pulse 3 conditions.

It can be seen across all electrodes that machining time is reduced with increase in pulse energy except in case of D_{NC-3} & D_{NC-8} running at pulse 3 in which machining time is very negligibly greater than pulse 2. At higher pulse energy debris removal is not sufficient which can be a contributing factor to the limit of this decrease in machining time for pulse 3. Significant reduction in machining time can be observed from pulse 1 to pulse 2 across all electrodes without any exception. Therefore, it can be said with certainty that by increasing pulse current I_p machining time can be reduced. D_{NC-6} maintained lowest machining time for pulse 1 & pulse 3. Lowest machining time for pulse 2 was in D_{NC-8} design. For pulse 2 machining condition D_{NC-8} reduces the machining time up to 62.61% when compared with conventional tool electrode.

3.2. Material removal rate

MRR for all electrode designs for 3 pulses was measured. It was observed that the highest MRR 36.742 mm^3/min was achieved using D_{NC-6} electrode applying pulse 3. For this pulse, this design recorded 69.33% increase in MRR. In addition, D_{NC-6} electrode design also indicated peak MRR for pulse 1 that is 9.579 mm^3/min, closely followed by D_{NC-8} design with 9.266 mm^3/min, trailed by D_{NC-3} design with a MRR of 9.172 mm^3/min.

For pulse 2, D_{NC-8} design with 34.496 mm^3/min MRR stayed the highest, closely followed by D_{NC-3} design with 33.920 mm^3/min. It can be observed that the least performing electrodes include the D_{NC-1} design with 6.189 mm^3/min and 20.773 mm^3/min MRR for pulse 1 and 3 respectively. MRR of this design is even lower compared to conventional design. MRR increased successively from pulse 1 to pulse 3 in all designs except D_{NC-3} & D_{NC-8} in which MRR of Pulse 3 is slightly lower than pulse 2 (Fig. 7).

For the validation of the significance of the improvements observed in MRR, critical experiments were repeated ($n = 3$) for D_C and D_{NC-6} at all 3 current levels and for D_{NC-8} at $I = 5A$. Table 9 presents the mean MRR values with standard deviation and 95% confidence intervals (CI). The narrow CIs indicate high repeatability of the experimental results. The replicated data confirms the trends observed in single runs, with D_{NC-6} consistently outperforming D_C across all current levels.

One-way ANOVA was performed to compare the mean MRR of D_C, D_{NC-6}, and D_{NC-8} at the 5A condition (Table 10). The analysis revealed difference between electrode designs ($F(2,6) = 13192.2$, $p < 0.0001$, $\eta^2 = 0.9998$). The effect size of 0.9998 indicates that electrode design accounts for 99.98% of the variance in MRR, confirming that the observed differences are not due to random variation but reflect legitimate performance differences between designs.

Tukey HSD post-hoc analysis with Bonferroni correction was used to identify which specific electrode pairs differ significantly (Table 11). D_{NC-8} showed the highest improvement over D_C at 164.0% (95% CI 160.9-167.1%, adjusted $p < 0.001$), followed by D_{NC-6} at 79.5% (95% CI 76.4-82.6%, adjusted $p < 0.001$). The comparison between D_{NC-8} and D_{NC-6} was also significant (47.1% improvement, adjusted $p < 0.001$), confirming that the 10° relief angle design provides superior performance at the 5A condition.

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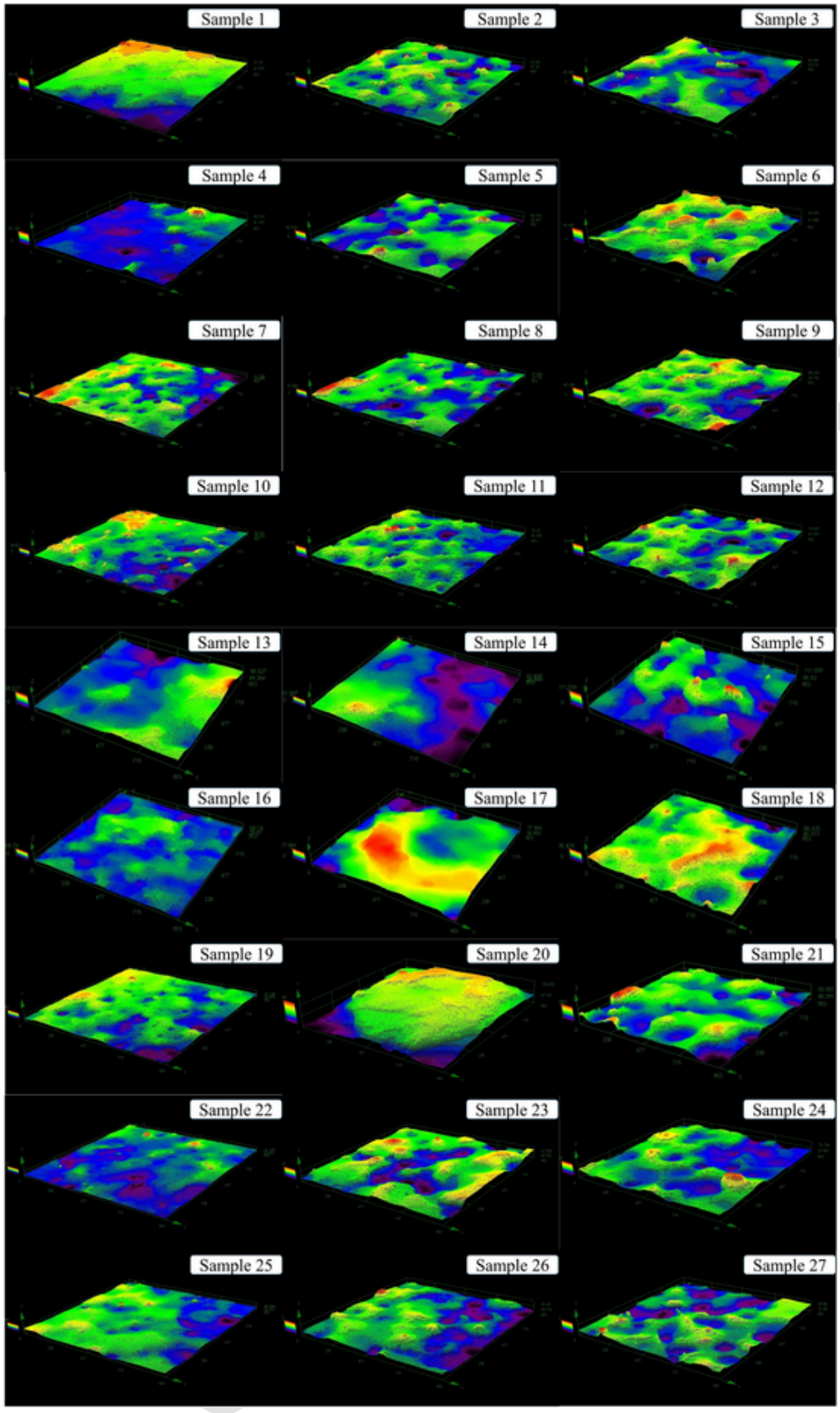


Fig. 15. Surface Roughness Three-Dimensional Plots of 27 machined samples with different electrodes, measurements in all three axes are in micrometers (μm).

PROOF

Table 12
Surface Micrographs of all machined samples.

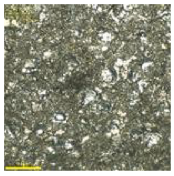
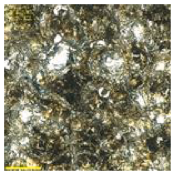
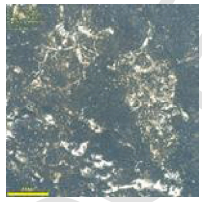
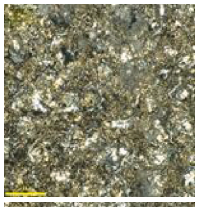
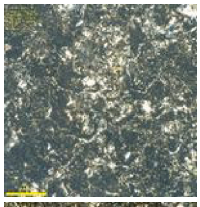
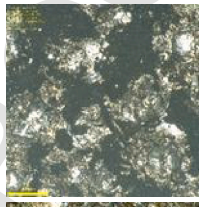
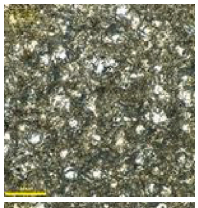
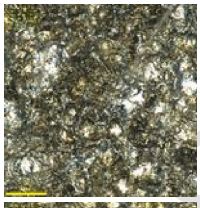
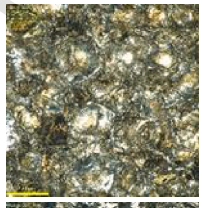
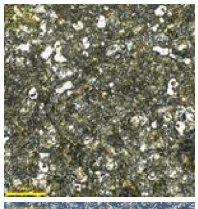
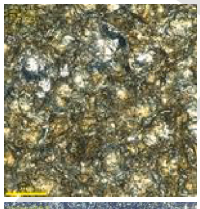
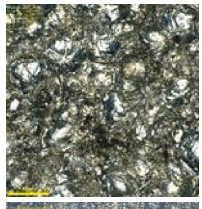
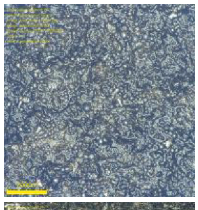
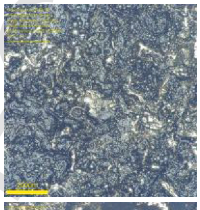
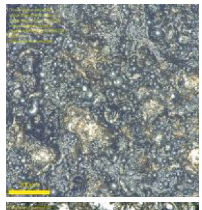
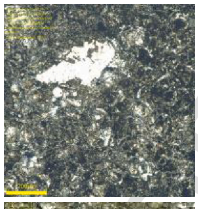
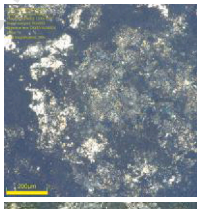
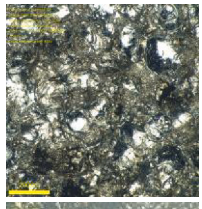
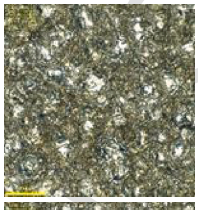
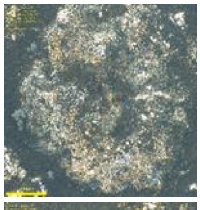
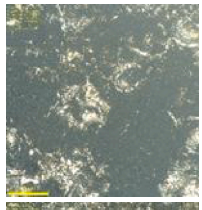
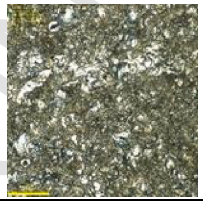
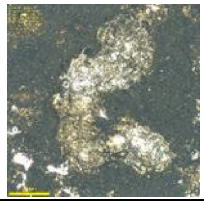
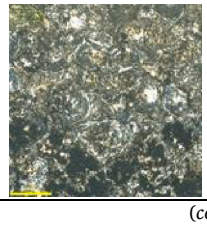
Electrode Design	Pulse 1	Pulse 2	Pulse 3
D_C			
D_{NC-1}			
D_{NC-2}			
D_{NC-3}			
D_{NC-4}			
D_{NC-5}			
D_{NC-6}			
D_{NC-7}			

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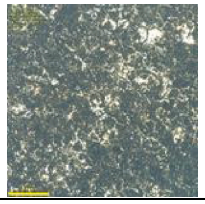
Table 12 (*continued*)

Electrode Design	Pulse 1	Pulse 2	Pulse 3
DNC-8			

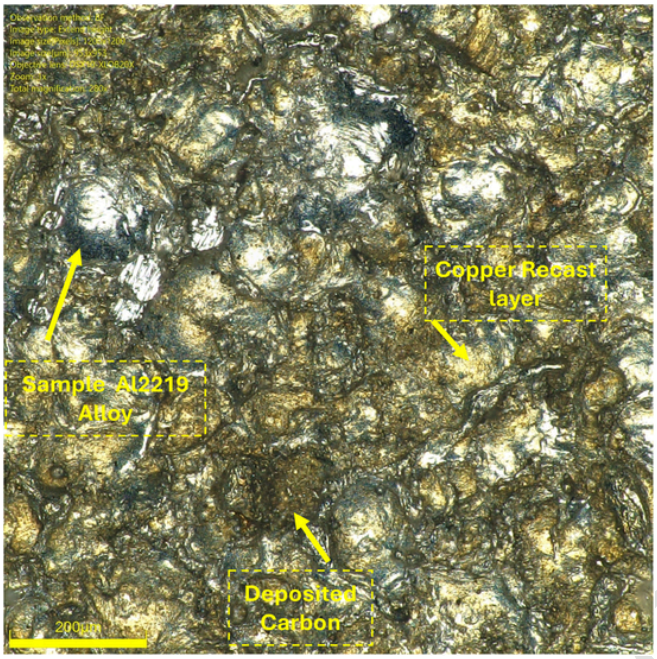


Fig. 16. Surface micrograph (280x magnification) of Sample 11 after die-sinking EDM using a modified electrode with 4 mm Dia.

3.3. Tool wear rate

TWR was calculated with mass difference of electrode before and after machining and density of Cu as TWR calculated from dimensional approach may lead to ambiguous results due to non-uniform wear and pitting on tool electrode surface that is challenging to measure using conventional metrology techniques.

For the presented system, TWR successively increased from pulse 1 to pulse 3 in all designs. TWR is thus highly dependent on I_p. Increase in pulse current leads to increased wear of electrode. TWR for pulse 2 and pulse 3 are recorded highest in DNC-3 design i.e. 0.162 mm^3/min & 0.260 mm^3/min respectively. For pulse 1 highest TWR is recorded in DNC-5 design i.e. 0.042 mm^3/min. In terms of tool wear rate D_c remains lowest for pulse 2 & 3. Lowest TWR in pulse 1 is recorded in DNC-1. Therefore, it can be concluded that conventional design leads to lower tool wear rates at the cost of slower machining process (Fig. 8).

The conventional electrode design (Dc) features a cylindrical geometry with a large continuous cylindrical area. This large cylindrical area spreads the discharge energy upon a greater volume of tool electrode material that reduces localized energy/current density and therefore minimizes volumetric tool wear. But the same geometry limits the flow of dielectric fluid and debris removal which leads to slower machining. Non-conventional electrodes on the other hand have channels for the flow of dielectric fluid therefore reducing machining time but at the same time these DNC designs have reduced volume that concentrates thermal load which in turn increases localized wear when compared to the conventional design.

3.4. Overcut

Overcut increased in all designs from pulse 1 to pulse 3. In this performance parameter, all electrodes demonstrated a similar behavior. This leads to the conclusion that overcut is highly dependent on pulse energy rather than conventional or non-conventional design. Low energy pulse although takes more machining time but provides higher accuracy. Highest overcut was measured 0.22 mm while lowest overcut was 0.12 mm (Fig. 9).

3.5. Surface roughness

Both two-dimensional average roughness Ra and three-dimensional average roughness Sa were measured for the presented cases. Ra was measured for cylindrical as well as bottom surface of blind holes in all samples. Sa was measured for bottom surface of blind holes.

Ra provides assessment of surface texture along a single path while Sa measure average deviation over a specified three-dimensional area. Ra provides quick measurement while Sa provides thorough information about surface texture. Each of Ra and Sa of cylindrical and bottom surface of hokes are plotted as follows (Fig. 10).

3.6. Discussion

From the presented results, it was observed that the lowest Ra 2.064 μm in cylindrical area of the blind hole was achieved using DNC-2 electrode in pulse 2. Also, lowest Ra 0.492 μm in bottom area was archived using DNC-5 electrode applying pulse 1 (Fig. 11). Furthermore, the lowest Sa 1.759 μm is measured in DNC-7 electrode using pulse 1. It was also observed that Ra generally increased with higher pulse energy. Using DNC-3 electrode bottom surface was rougher in pulse 2 as compared to pulse 3. Highest surface roughness was recorded in DNC-1 electrode as Sa value of 9.371 μm. Bottom area is rougher than cylindrical area due to accumulation of debris that could not be flushed and high energy sparks over this debris lead to rougher surface. Sa provided supplementary information about three-dimensional consistency of the surface texture, specifically for surfaces with irregular crater distributions. Electrode with relief angle 10-degree DNC-8 performed particularly well in this performance parameter.

DNC-6 electrode design with grooves is the lightest among all designs and Conventional design is the heaviest. Highest MRR 36.742 mm^3/min was accomplished using DNC-6 electrode in pulse 3 followed by DNC-8, DNC-3, DNC-7 respectively. Weight removal strategy from the tool electrode without altering face diameter seemingly increase material removal rate at first glance due to improved circulation of dielectric. But DNC-3 performed better than DNC-7 although it had more weight. So, it can be concluded that Geometrical design along with weight reduction played crucial role in determining MRR and other performance variables. That is why different designs were experimentally tested (Fig. 12).

Fig. 13 shows the trend of MRR and TWR for two groups of electrode design. MRR increased along with TWR when more light weighted electrodes were employed. Remarkable decrease in TWR was recorded for DNC-6 without compromising its MRR due to excep-

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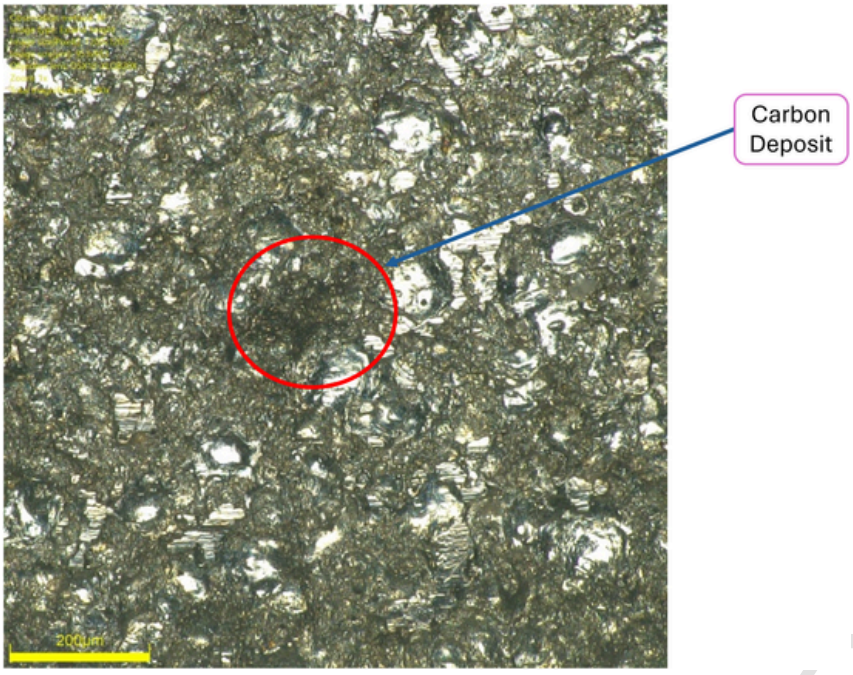

Fig. 17. Surface micrograph (280x magnification) of Sample 1 after die-sinking EDM using a conventional cylindrical electrode.

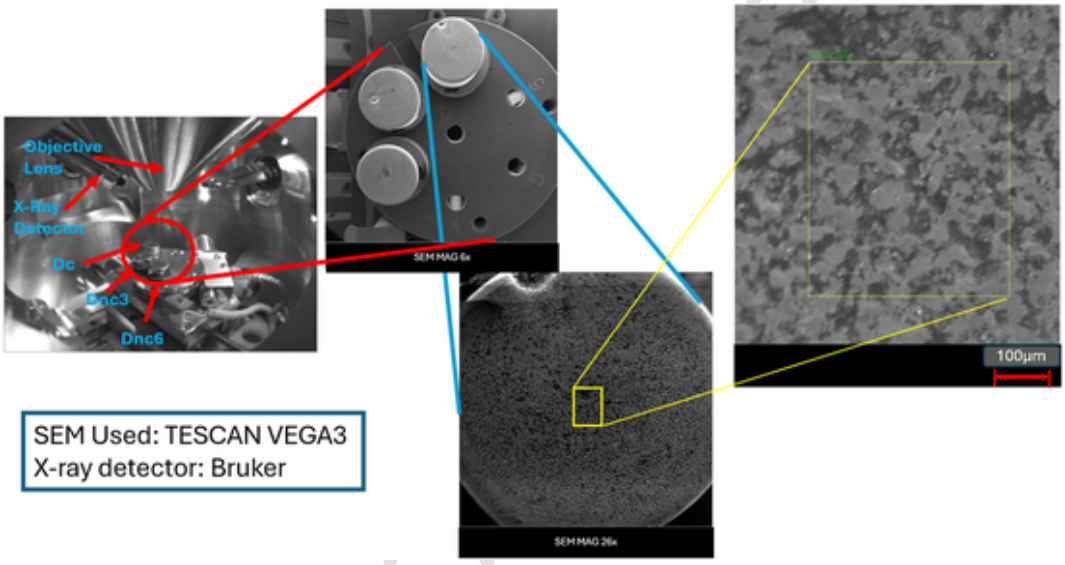

Fig. 18. Sem of Dc, Dnc-3 and Dnc-6.

tional flushing. This design gave better path for bubble and debris removal which resulted in higher MRR and lower TWR.

Following the machining experiments, the base surface of each of the machined samples were examined using the Olympus DSX1000 digital microscope (Fig. 14). This instrument was selected because its non-contact optical surface profiling capability allowed accurate roughness measurement at the hole base of machined samples. Bottom of machined holes is a region that is geometrically inaccessible to conventional stylus-based instruments such as perthometer.

For each specimen, a three-dimensional topographic map of the central region of the hole bottom was reconstructed by acquiring a stack of images at incremental focal planes. It was done by giving focal range of highest and lowest point to the operating system, which then automatically acquire stack of image at different heights and compile these images to form a three-dimensional mapping of the surface. From this reconstruction, both the two-dimensional arithmetic mean roughness (Ra) and the three-dimensional areal roughness (Sa) were computed using the microscope's integrated analysis software.

Three-dimensional wireframe surface plot was obtained from all machined samples (Fig. 15). The wireframe reconstruction clearly illustrates the distribution of peaks and valleys generated by the EDM process. The color-coded height scale indicates the vertical extent of surface irregularities, with warmer colors (red and yellow) representing elevated peaks and cooler colors (blue and green) denoting deeper craters. The wireframe plot also reveals the spatial periodicity of the discharge craters, which reflects the influence of the electrode design and the stability of sparking during machining. The numerical values displayed along the axes correspond to the lateral dimensions in micrometers (µm) and the vertical amplitude of the surface texture that too in micrometers (µm), providing a quantitative visual representation of the EDM-induced topography at the blind hole base.

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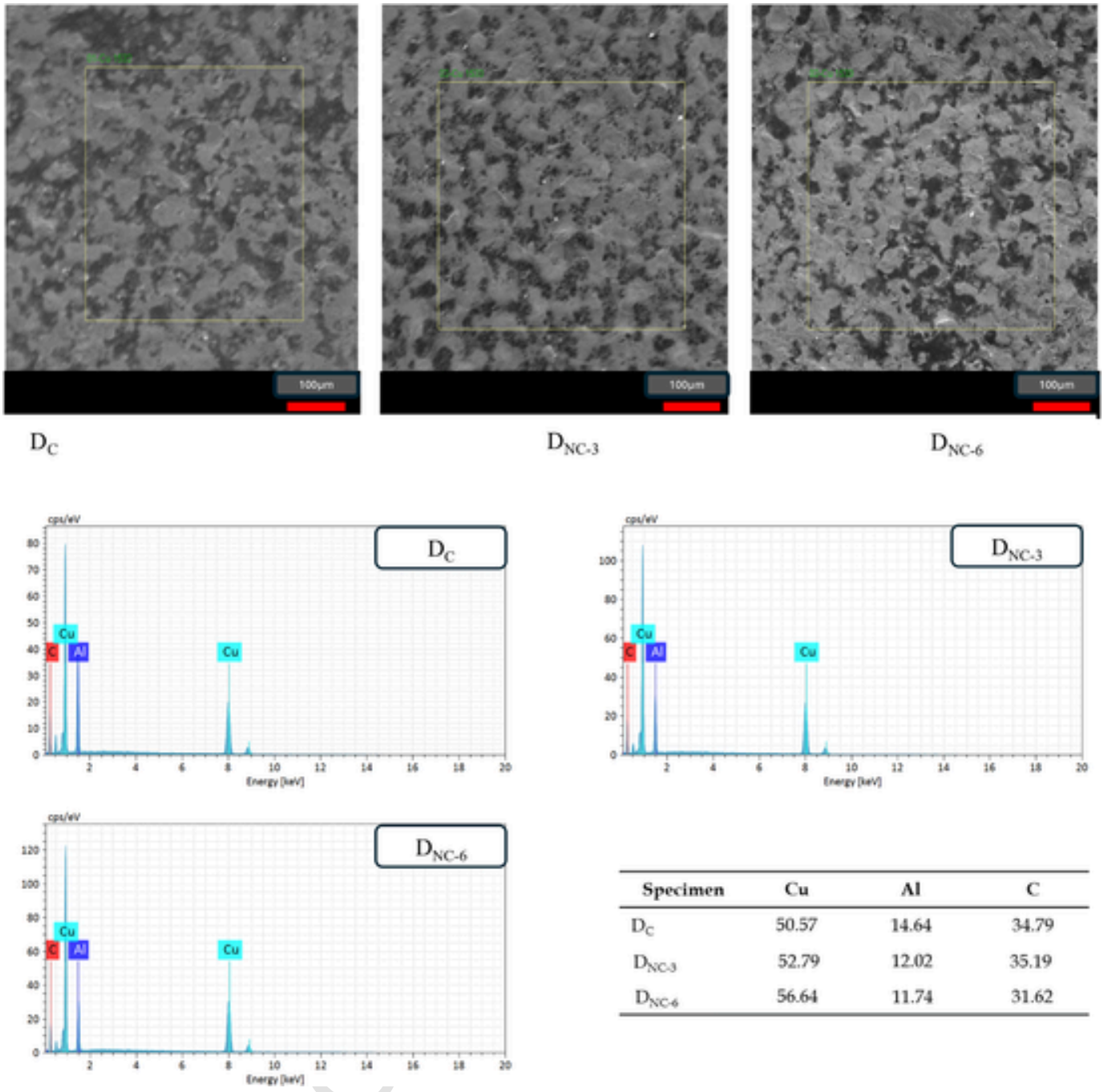

Specimen	Cu	Al	C
D_C	50.57	14.64	34.79
D_{NC-3}	52.79	12.02	35.19
D_{NC-6}	56.64	11.74	31.62

Fig. 19. EDX of electrodes D_C, D_{NC-3} and D_{NC-6}.

Table 12 gives surface micrographs of all 27 samples. These micrographs of bottom surface were taken. Table shows that debris of carbon is greater in pulse 2 & 3 samples due to higher pulse energy more dielectric is decomposed into carbon and deposited at the bottom crevices that require intense post machining process.

At 280x magnification (Fig. 16), the surface of the Al2219 specimen number 11, machined with the modified electrode (4 mm dia cuts) under the medium-current pulse condition (5 A) exhibits typical EDM induced features. The surface is characterized by overlapping craters of varying size and depth; interspersed with resolidified molten material that forms an irregular recast layer. Notably, several discrete, bright-contrast regions are visible across the micrograph; these are attributed to the deposition of copper originating from the electrode. The medium current setting (5 A) generates sufficient discharge energy to melt and transfer electrode material onto the workpiece surface without causing excessive crater coalescence. The presence of copper deposits is confirmed by their distinct optical reflectivity under the microscope; appearing as localized patches or globular features embedded within the aluminum matrix. These copper-rich regions are indicative of the material migration phenomenon intrinsic to die-sinking EDM, where the electrode not only erodes but also contributes to surface alloying. The overall surface morphology suggests stable sparking conditions with moderate thermal input, leading to a relatively uniform distribution of craters and deposited material across the observed field of view that is approximately 1 mm × 1 mm.

Fig. 17 presents the surface of the machined sample using a conventional cylindrical electrode under a low current pulse condition (3 A). The surface is covered by a continuous dark patched area at the

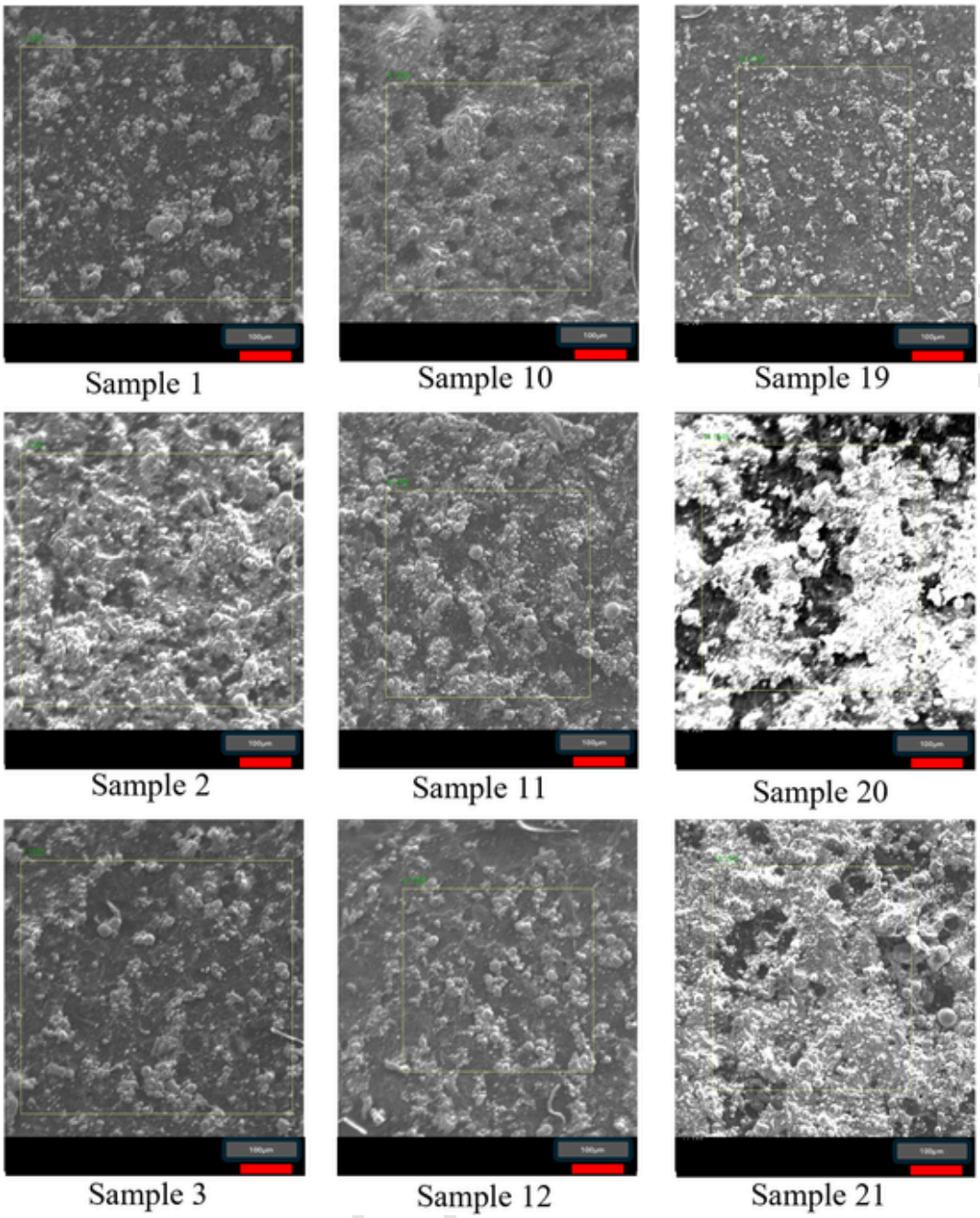

Fig. 20. EDX of workpiece samples developed using D_C, D_{NC-3} and D_{NC-6}.

center, continuous residue spanning approximately 260 μm in length. This black layer is attributed to the thermal decomposition of the hydrocarbon dielectric fluid during the electrical discharge process. This shows that flushing intensity is insufficient to completely expel the pyrolyzed carbon products from the machining gap. Consequently, these carbon-rich residues adhere to the workpiece surface, forming a patchy or continuous black film. Additionally, the low current condition may not help with the flushing efficiency. The presence of this residue indicates that while the conventional electrode with low current minimizes surface cratering and thermal damage, it also promotes the deposition of carbon on workpiece, which may affect subsequent surface quality and post-processing requirements.

The EDX analyses were performed after SEM analysis through TESCAN VEGA3 (Fig. 18). The analysis was conducted for the tools as well as the workpieces (Figs. 19 and 20 & Table 13), which provided quantitative insight into surface composition after EDM. Since Al2219 contains 5.8-6.8 wt% Cu as a major alloying element, along with 0.2-0.4 wt% Mn, 0.1-0.25 wt% Zr, and up to 0.3 wt% Fe, the elemental readings on machined surfaces must be interpreted relative to this baseline composition. (Table 13). It was observed that the carbon content on all machined surfaces was remarkably high, ranging from 71.01% for Sample 10 (DNC-3 at 3A) to 75.53% for Sample 12 (DNC-3 at 7A), with an average of approximately 72.5% across all samples measured. It was also observed that for sample produced through higher currents (Sample 2, 3, 11, 12 and 20), carbon deposition increased. The only exception to this was found in sample 11 which had only marginal rise to 71.58% deposition to 71.28% of 3A of sample 19. Sample 12 (DNC-3 at 7A) showed the highest carbon content at 75.53%, while Sample 1 (conventional electrode at 3A) showed the lowest at 71.03%. The conventional electrode, even though had poor flushing characteristics, pro-

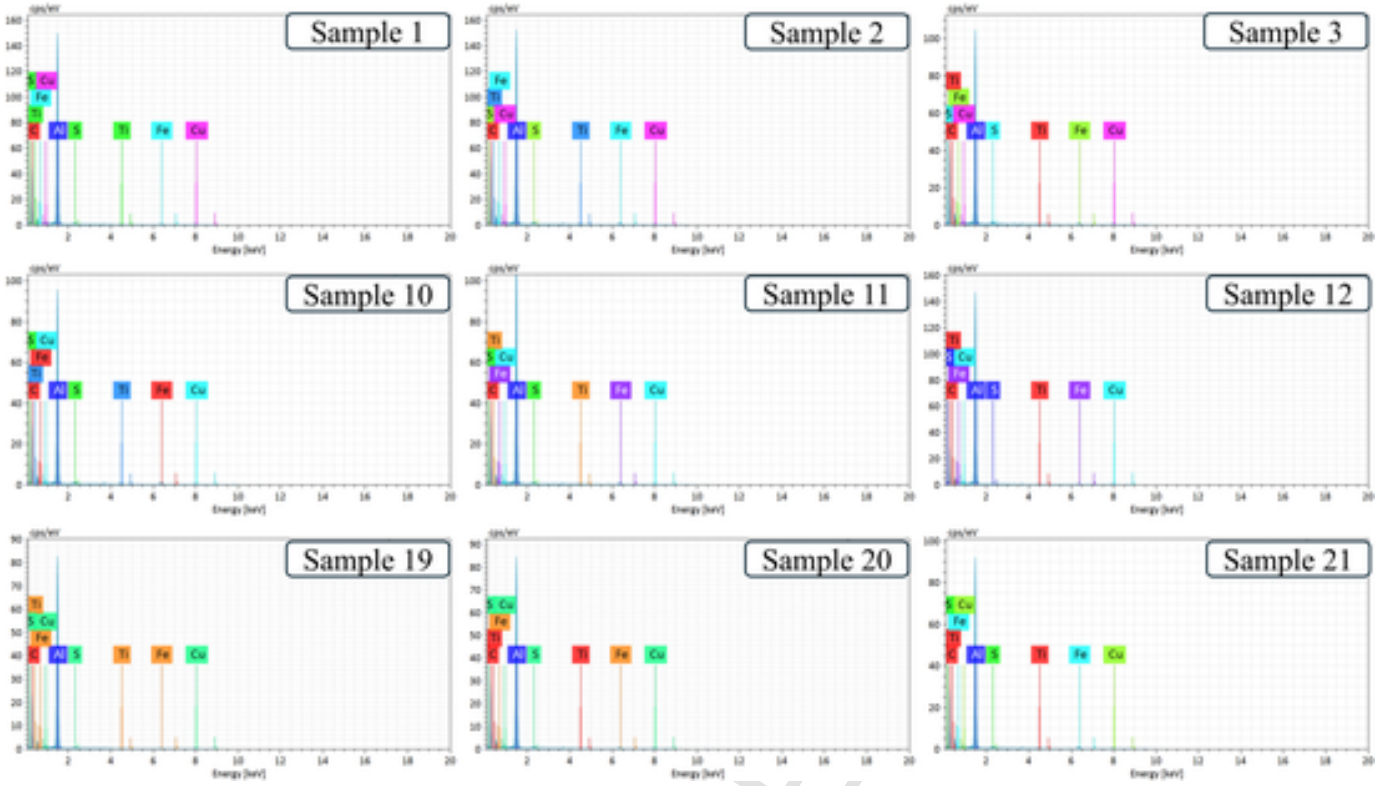

Fig. 20. (*continued*)

Table 13
Quantitative EDX mapping of the samples developed using Dc, Dnc-3 and Dnc-6.

Tool Utilized	Specimen	Carbon	Aluminum	Sulfur	Titanium	Iron	Copper
Dc	Sample 1	71.03	26.41	0.64	0.27	0.55	1.10
	Sample 2	73.14	21.87	0.66	0.87	1.54	1.92
	Sample 3	74.23	21.25	0.66	0.54	1.32	2.00
Dnc-3	Sample 10	71.01	23.18	0.58	1.20	2.00	2.03
	Sample 11	71.34	24.67	0.39	0.58	1.29	1.73
	Sample 12	75.53	21.80	0.24	0.32	0.66	1.45
Dnc-6	Sample 19	71.28	25.13	0.44	0.25	0.81	2.09
	Sample 20	73.55	22.29	0.48	0.57	1.26	1.85
	Sample 21	71.59	24.84	0.41	0.50	1.13	1.53

duced the lowest carbon content under low current conditions due to lower discharge energy at 3A that yields less dielectric decomposition and therefore less carbon to deposit. At higher energy, carbon decomposition increased due to more intense discharge activity. Modified electrodes, although better flushing, showed slightly higher carbon content in few cases that can be explained by the fact that enhanced flushing removes debris more effectively from the gap that exposed fresh dielectric to later discharges bringing more carbon to deposit.

The copper content on machined surfaces ranged from 1.10% for Sample 1 to 2.09% for Sample 19. When compared to the 5.8-6.8 wt% copper in the base Al2219 alloy, these values show substantial copper reduction in the recast layer. This reduction occurs because although copper has a higher boiling point than aluminum, copper vaporizes more promptly under decreased pressure conditions in the discharge gap. During EDM cycle, copper favorably vaporizes from the molten pool leaving behind a surface layer depleted in copper. Modified electrodes machined samples showed higher copper content compared to conventional electrode samples. The average copper content for non-conventional electrodes samples was 1.83%, while conventional electrode samples were 1.73% because with their improved flushing localized overheating and copper vaporization was reduced.

Sulfur content 0.24% to 0.66%, likely originating from dielectric fluid. Titanium content 0.25% to 1.20%, and iron content ranged from 0.55% to 2.00% may originate from minor impurities in the workpiece material or fixture materials. The presence of iron may indicate material transfer from the machine tool or electrode holder. These trace elements do not affect machining performance but indicate presence of complex material transfer process.

Line scans were carried out on samples 3, 12, and 21 (Fig. 21). These three samples were selected because they were machined at the highest current level (7 A). Higher current produces greater discharge energy and therefore the most favorable conditions for carbon diffusion. If no diffusion is observed under these conditions, it is unlikely to occur at lower currents. EDX testing of the samples (workpiece processed surfaces) confirmed the presence of carbon of machined surface from spark erosion. To assess carbon diffusion into the substrate, EDX-line scan analysis was performed across cut and polished cross-section of samples machined by EDM.

As observed in the line scan profile of sample 3 (Fig. 21a), carbon signal noticeably higher only within first 20-30 μm from the surface which corresponds to recast layer. Beyond this layer carbon peak rapidly drops and signal remains low across the scan length with slight noise observed intermittently. This abrupt drop without gradual concentration indicates that carbon has not diffused into the substrate into the base metal beyond recast layer. These results are consistent with EDM mechanism in which carbon from dielectric fluid gets entrapped into the molten pool, it is then trapped within recast layer upon rapid solidification.

Line Scan graphs of sample 3 and sample 12 both show the same trend. Carbon signal is evidently higher in the initial 25 μm and then smooths to a low stable baseline for the remainder length in both samples. This is evidence of carbon rich layer that is confined to recast layer and no diffusion into base metal. In sample 3, aluminum signal starts at low edge the rise sharply and remains high for the rest of the line scan. This is entirely plausible as base metal is Al alloy. The initial low signal of Al in sample 3 indicates non-Al rich surface layer. In sample 12 same

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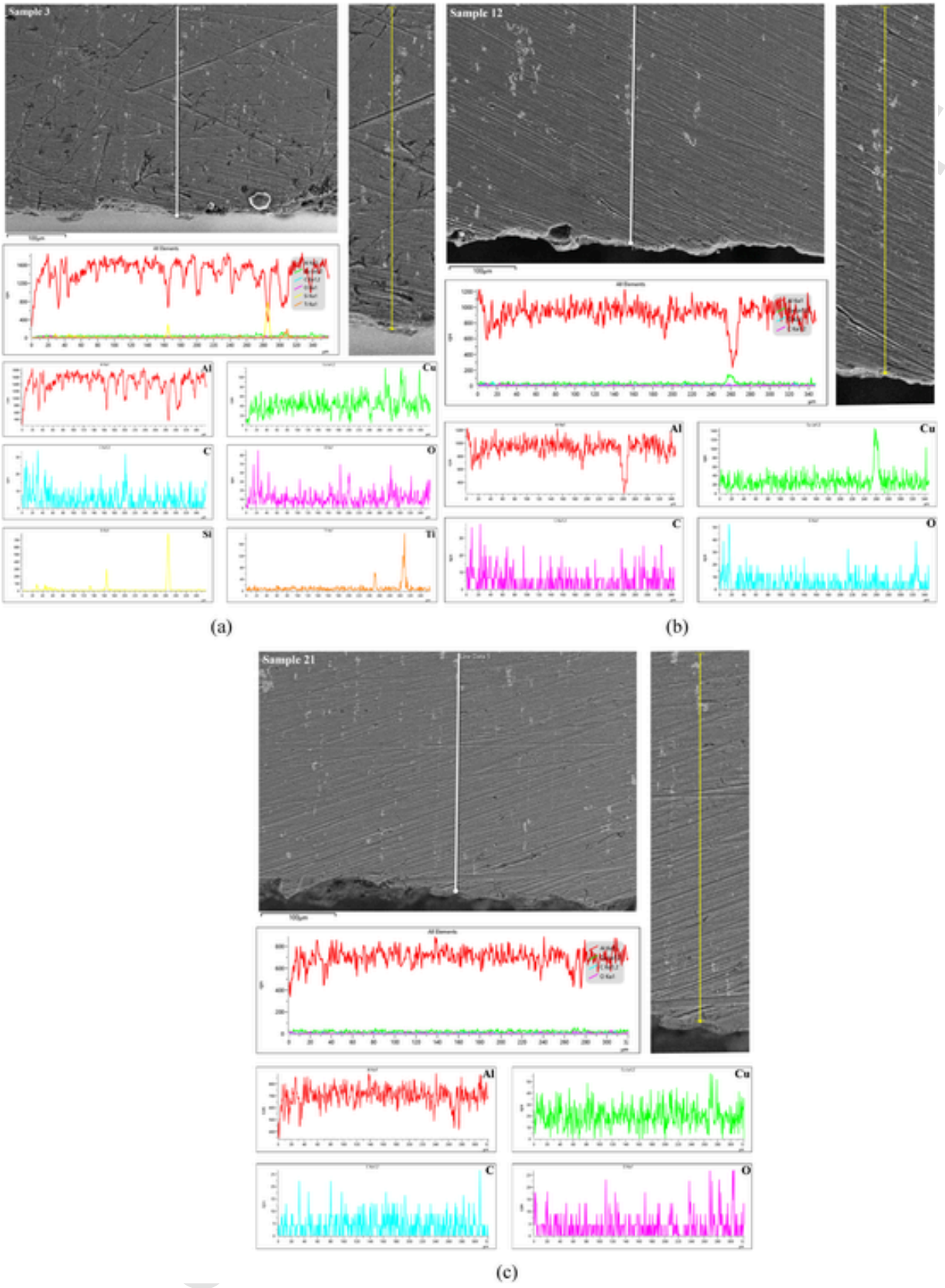

Fig. 21. EDX line scan of (a) sample 3 (DC, 7 A), (b) sample 12 (DNC-3, 7 A), and (c) sample 21 (DNC-6, 7 A). Carbon is high only in the first 20–30 μm (recast layer) for all 3 cases and then drops sharply, with no gradient into the base metal.

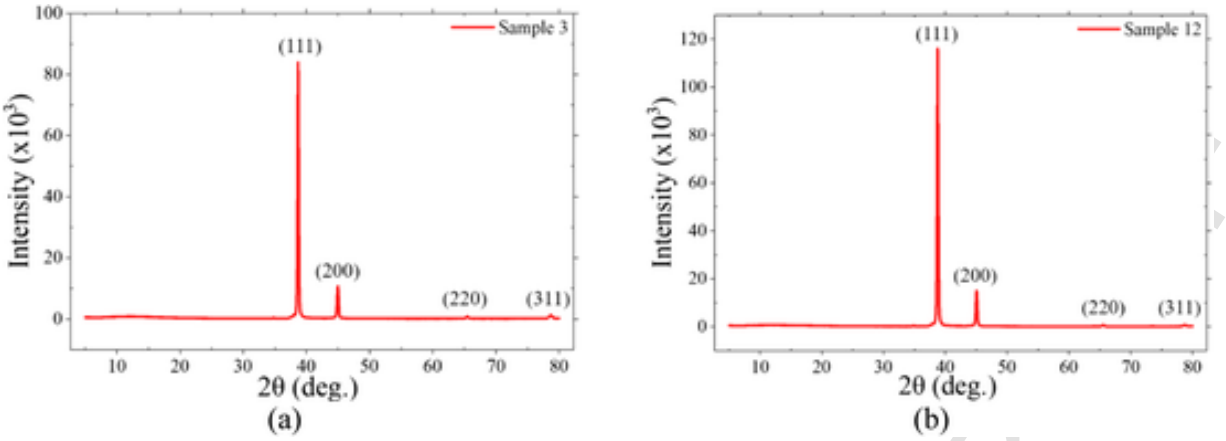

Fig. 22. XRD patterns of samples machined at 7 A using DC (sample 3) and DNC-3 (sample 12).

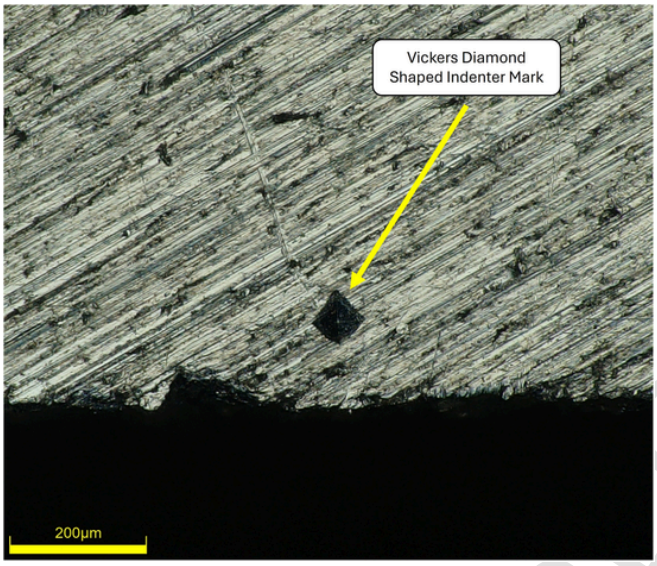

Fig. 23. Microhardness test on Ground and polished sample 3 on Heat affected Zone.

Table 14
Mean microhardness results of the 3 samples at 3 5 & 7A respectively.

Specimen	Current	Mean Hardness (HV)
Sample 1	3A	160.6
Sample 2	5A	161.8
Sample 3	7A	168.0

trend is found for aluminum except there is a notable dip at approximately 260 μm from the top layer. This dip is caused by a second phase particle in the scan path, not a compositional gradient. This phenomenon is confirmed by corresponding Cu spike at the same position. Copper signal remained stable in both sample 3 and sample 12, except that spike in sample 12 which is a Cu-rich intermetallic particle in this Aluminum alloy. Stable copper signal in both samples show it as bulk alloying element rather than surface contaminant like carbon. Oxygen signal appears in sample 3 and 12 that is slightly elevated near surface region then becomes low and noisy throughout the rest of the scan. This is consistent with mild surface oxidation. Sample 3 also shows traces of Si and Ti; signals of these trace elements are mostly very low and noisy, indicating trace presence.

In sample 21 aluminum signal starts low from 0 to 5 μm and then sharply rises to stable high. This trend has been noticed in previous sample. Copper signal consistently remains stable at constant level with slight noise. The existence of carbon and oxygen signal with cps below 60 for all samples as well as the absence of gradient of carbon from surface into the base metal suggests no carbon diffusion into the substrate.

X-ray diffraction (XRD) analysis was performed to determine the phase composition of the machined samples with EDM. XRD patterns for the Al2219 solid specimens processed under die-sinking EDM are presented in Fig. 22. Constructive interference on different crystallographic planes yielded varied diffraction intensities, representing the primary Face-Centered Cubic (FCC) α-Al matrix phase. Sharp high intensity peaks were recorded at 2Θ values of approximately 38.7°-38.8°, 45.02°, 65.4° and 78.6°. This indicates diffraction on the crystallographic planes (111), (200), (220) and (311) respectively (ICDD card no. 00-04-0787). No diffraction peaks corresponding to crystalline carbon forms such as aluminum carbide (Al_4C_3) or amorphous carbon forms developed because any carbon from hydrocarbon dielectric decomposition remained below the crystalline detection limit of XRD.

Struers Duramin Vicker hardness tester was used to measure microhardness (Fig. 23). Base metal exhibited an average hardness of 128.5 HV. Post EDM, at 3A, 5A, and 7A; Al-2219 showed average hardness of 160.6 HV, 161.8 HV, and 168.0 HV in samples 1, 2 and 3 respectively machined with conventional electrode (Table 14). This indicates the formation of a hardened recast layer on the machined surface. This represents a 25% to 31% increase in surface hardness compared to the base material. The increase in hardness is attributed to the formation of a recast layer on the machined surface. Rapid re-solidification of molten material produces residual stresses. The presence of carbon from dielectric decomposition and oxides formed during discharge also contributes to this hardening effect. The variation in hardness across current levels was also observed. At 3A, A current produced a surface hardness of 160.6 HV, which increased to 168.0 HV at 7A. This increase of 4.6% suggests that the current level influences surface hardness.

Considering microhardness and EDX data together, shows that 25-31% increase in hardness can be attributed to the formation of recast layer within heat affected zone. High carbon content in recast layer yielded hardening through solid solution strengthening. Although copper depletion softens this alloy, it is overweighed by the incorporation of carbon.

To assess the performance of these electrodes under the 3 different current levels, radar plots were developed (Fig. 24). At low energy setting (Fig. 24a), D_{NC-3} and D_{NC-8} exhibit largest enclosed area, showing the best overall balance among the performance parameters MRR, TWR, SR, Overcut and Machining time. When we move to medium current setting (Fig. 24b), D_{NC-3} and D_{NC-8} continue to show superior performance. But, at the highest current level (Fig. 24c), D_{NC-6} and D_{NC-7} overtake these and become the top performers. These geometries, due to their angular features, facilitate better flushing under intense spark-

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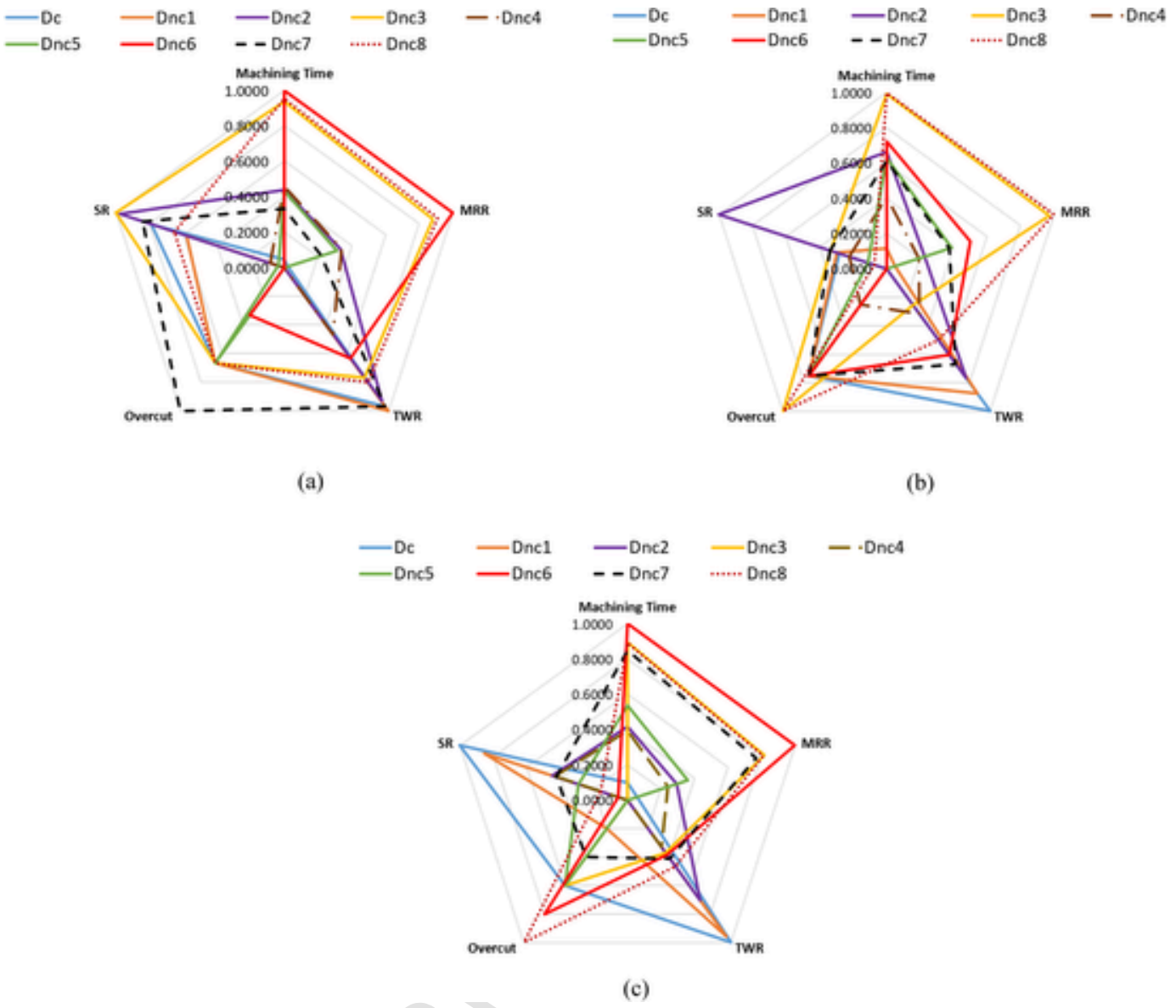

Fig. 24. Performance parameter radar plots for all currents of (a) Pulse 1 (b) Pulse 2, and (c) Pulse 3.

ing conditions. However, it has been observed that for most of the generalized current conditions, $D_{NC\text{-}3}$ and $D_{NC\text{-}8}$ are the recommended options.

4. Conclusion

This study investigated the influence of tool geometry on machining performance and surface integrity during die-sinking EDM of Al2219 across three current levels. Nine electrode designs were tested, including a conventional cylindrical electrode and eight modified geometries with repeated pattern features. The absolute highest MRR value of 36.431 ± 0.292 mm^3/min (n = 3) was achieved using $D_{NC\text{-}6}$ at 7A, representing a 68.6% improvement over the conventional electrode, while also reducing machining time to 22.10 min. In addition, the highest improvement by percentage in MRR was achieved by DNC-8 at 5A, with a 164.0% increase (95% CI 160.9–167.1%, adjusted $p < 0.001$) over the conventional electrode. $D_{NC\text{-}8}$ demonstrated superior bottom surface finish at 3A with Ra of 1.204 μm and Sa of 2.342 μm. The lowest TWR was recorded for $D_{NC\text{-}1}$ at 3A (0.012 mm^3/min), though conventional electrode showed lower wear at higher currents. Overcut increased with current level across all designs, with $D_{NC\text{-}7}$ achieving the smallest overcut of 0.12 mm at 3A. Surface roughness generally increased with higher pulse energy, with bottom surfaces being rougher than cylindrical walls due to debris accumulation. After evaluating all parameters, $D_{NC\text{-}3}$ and $D_{NC\text{-}8}$ provide a balanced approach between MRR and surface finish with moderate TWR and low overcut. Line-scan EDX confirmed that carbon remained confined to the recast layer with no diffusion into the base metal, and XRD showed no new crystalline phases such as Al_4C_3. Future work may explore higher aspect ratios, constant-mass controls, alternative electrode and workpiece materials, and CFD modeling of dielectric and debris flow for different electrode designs.

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CRediT authorship contribution statement

Muhammad Tayyab: Investigation, Visualization, Writing – original draft, Writing – review & editing. **Syed Maaz Hasan:** Conceptualization, Data curation, Formal analysis, Writing – original draft, Writing

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– review & editing, Supervision, Project administration. **Emad Uddin:** Conceptualization, Writing – review & editing, Supervision. **Sadaqat Ali:** Conceptualization, Writing – review & editing, Supervision.

Consent to participate

Not applicable. No participants were involved in this study.

Ethical considerations

Not applicable. No ethical approval was required for this study.

Consent for publication

Not applicable. No consent for publication was required.

Data availability

The data that support the findings of this study are available from the corresponding author upon request.

Funding statement

No funding was received for this study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Xu M, Yang Y, Chu X, Cao L, Li J, Li C, et al. Advances in the experiment of ultrasonic and electrical discharge-assisted machining of titanium alloys: a systematic review. J Mater Res Technol 2026;42:4165–85. https://doi.org/10.1016/j.jmrt.2026.04.086.

[2] Baroi BK, Jagadish Patowari PK. A review on sustainability, health, and safety issues of electrical discharge machining. J Braz Soc Mech Sci Eng 2022;44:59. https://doi.org/10.1007/s40430-021-03351-4.

[3] Hasan MM, Saleh T, Sophian A, Rahman MA, Huang T, Mohamed Ali MS. Experimental modeling techniques in electrical discharge machining (EDM): a review. Int J Adv Manuf Technol 2023;127:2125–50. https://doi.org/10.1007/s00170-023-11603-x.

[4] Mohd Abbas N, Solomon DG, Fuad Bahari Md. A review on current research trends in electrical discharge machining (EDM). Int J Mach Tool Manufact 2007;47:1214–28. https://doi.org/10.1016/j.ijmachtools.2006.08.026.

[5] Singh N, Saluja RK, Gahlot NK, Saini GS. Existing development techniques and machining in MMCs for automotive and aeronautical industries. Industrial composites, optimization methods, and tribological considerations. CRC Press; 2026.

[6] Li H, Wang C, Yu Q, Cao Z, Ma Y, Fang X. Enhancing the bottom surface flatness of blind holes in electrochemical drilling by optimizing the electric and flow field with discretized electrode. Int J Adv Manuf Technol 2024;135:1617–31. https://doi.org/10.1007/s00170-024-14615-3.

[7] Ayesta I, Flaño O, Izquierdo B, Sanchez JA, Plaza S. Experimental study on debris evacuation during slot EDMing. Proced CIRP 2016;42:6–11. https://doi.org/10.1016/j.procir.2016.02.174.

[8] Tanjilul M, Ahmed A, Kumar AS, Rahman M. A study on EDM debris particle size and flushing mechanism for efficient debris removal in EDM-drilling of Inconel 718. J Mater Process Technol 2018;255:263–74. https://doi.org/10.1016/j.jmatprotec.2017.12.016.

[9] Najati Ilkhchi R, Shabgard M, Kabirinia F. Numerical studying and experimental investigation: effect of Reynolds number on performance measures of EDM with high speed flushing. J Manuf Process 2019;48:228–35. https://doi.org/10.1016/j.jmapro.2019.10.035.

[10] Xu Y, Ding Y, Xie W, Chen K, Yu Z, Liu H, et al. Precision fabrication of high-aspect-ratio micro-holes in 40 mm-thick TC4 alloy by coaxial electrolyte-enveloped laser-electrochemical hybrid machining. J Mater Res Technol 2026;42:5323–44. https://doi.org/10.1016/j.jmrt.2026.04.076.

[11] Meena VK, Azad MS, Mitra S. Effect of flushing condition on deep hole micro-EDM drilling. IJMMM 2012;12:308. https://doi.org/10.1504/IJMMM.2012.050429.

[12] Sohani MS, Gaitonde VN, Siddeswarappa B, Deshpande AS. Investigations into the effect of tool shapes with size factor consideration in sink electrical discharge machining (EDM) process. Int J Adv Manuf Technol 2009;45:1131–45. https://doi.org/10.1007/s00170-009-2044-5.

[13] Plaza S, Sanchez JA, Perez E, Gil R, Izquierdo B, Ortega N, et al. Experimental study on micro EDM-drilling of Ti6Al4V using helical electrode. Precis Eng 2014;38:821–7. https://doi.org/10.1016/j.precisioneng.2014.04.010.

[14] Jacob KS, Samuel GL. Effect of micro double helical grooved tools on performance of electric discharge drilling of Ti-6Al-4V. Proc IME B J Eng Manufact 2021;235:1832–47. https://doi.org/10.1177/0954405421995605.

[15] Kumar R, Singh I. Blind hole fabrication in aerospace material Ti6Al4V using electric discharge drilling: a tool design approach. J Mater Eng Perform 2021;30:8677–85. https://doi.org/10.1007/s11665-021-06052-0.

[16] Wu B, Lei J, Lu J, Cheng R, Xu B, Cheng T, et al. Die-sinking micro electrical discharge machining of deep narrow groove arrays using self-sharpening sandwich electrodes. J Mater Res Technol 2026;43:3536–48. https://doi.org/10.1016/j.jmrt.2026.07.005.

[17] Goiogana M, Elkaseer A. Self-Flushing in EDM drilling of Ti6Al4V using rotating shaped electrodes. Materials 2019;12:989. https://doi.org/10.3390/ma12060989.

[18] Ahmed N. Die-sinking EDM performance under the effect of tool designs and tool materials. Int J Interact Des Manuf 2025;19:4951–72. https://doi.org/10.1007/s12008-024-02108-2.

[19] Ahmed N. Machining and wear rates in EDM of D2 steel: a comparative study of electrode designs and materials. J Mater Res Technol 2024;30:1978–91. https://doi.org/10.1016/j.jmrt.2024.03.242.

[20] Hirao A, Gotoh H, Tani T. Effect of electrode shape on high aspect ratio deep hole drilling by EDM. Proced CIRP 2022;113:262–6. https://doi.org/10.1016/j.procir.2022.09.156.

[21] Huang T-W, Sheu D-Y. High aspect ratio of micro hole drilling by Micro-EDM with different cross-section shape micro tools for flushing process. Proced CIRP 2020;95:550–3. https://doi.org/10.1016/j.procir.2020.01.161.

[22] Wu B, Lei J, Lu J, Wu Z, Xu B, Cheng T, et al. Exploration of morphological evolution of surface array microstructures in die-sinking micro-EDM with composite material electrodes. J Mater Res Technol 2026;42:75–90. https://doi.org/10.1016/j.jmrt.2026.03.090.

[23] Khanra AK, Sarkar BR, Bhattacharya B, Pathak LC, Godkhindi MM. Performance of ZrB2–Cu composite as an EDM electrode. J Mater Process Technol 2007;183:122–6. https://doi.org/10.1016/j.jmatprotec.2006.09.034.

[24] Faroze F, Srivastava V, Batish A. An investigation on the feasibility and performance analysis of EDM electrodes produced through rapid tooling route. Progr Addit Manuf 2026;11:2969–90. https://doi.org/10.1007/s40964-025-01506-6.

[25] S NJ, S MV, A BB. Investigating FDM electrodes for EDM machining of mild steel: performance evaluation. SSRG-IJME 2024;11:123–30. https://doi.org/10.14445/23488360/IJME-V11I6P114.

[26] Equbal A, Equbal MdI, Sood AK. An investigation on the feasibility of fused deposition modelling process in EDM electrode manufacturing. CIRP J Manuf Sci Technol 2019;26:10–25. https://doi.org/10.1016/j.cirpj.2019.07.001.

[27] Çakıroğlu R, Günay M. Comprehensive analysis of material removal rate, tool wear and surface roughness in electrical discharge turning of L2 tool steel. J Mater Res Technol 2020;9:7305–17. https://doi.org/10.1016/j.jmrt.2020.04.060.

[28] Singh NK, Singh Y, Sharma A, Prasad R. Experimental investigation of flushing approaches on EDM machinability during machining of titanium alloy. Mater Today Proc 2021;38:139–45. https://doi.org/10.1016/j.matpr.2020.06.165.

[29] Mao X, Wu G, Tran M, Yi S, Ding S. Electrical discharge drilling of blind holes with injection flushing dielectric and stepped electrodes. Int J Adv Manuf Technol 2024;132:495–511. https://doi.org/10.1007/s00170-024-13396-z.

[30] Sahoo R, Kumar D, Singh NK, Bajpai V. Fabrication of micro-hole using novel Maglev EDM. Journal of Micromanufacturing 2023;6:141–50. https://doi.org/10.1177/25165984221129449.

[31] Gao M, Liu M, Han J, Zhang Q. Investigation of machining characteristics in electrical discharge machining using a slotted electrode with internal flushing. Micromachines 2023;14:1989. https://doi.org/10.3390/mi14111989.

[32] Machno M, Trajer M, Czeszkiewicz A, Żurawka P. A Study on the channel and gap flow simulation for electrical discharge micro-drilling of inconel 718 superalloy. Adv Sci Technol Res J 2024;18:159–76. https://doi.org/10.12913/22998624/183998.

[33] Wang C, Sasaki T, Hirao A. Bubble behavior of single-pulse discharge in EDM. J Manuf Process 2024;126:60–73. https://doi.org/10.1016/j.jmapro.2024.07.106.

[34] Pratap S, Upadhyay A, Sharma V, Datta S, Badar MA. An analysis to enhance the machining performance of micro-EDM for drilling of blind micro-hole using ANN. Int J Adv Manuf Technol 2023;129:2551–61. https://doi.org/10.1007/s00170-023-12476-w.

[35] Wang Y, Fan L, Shi J, Dong Y, Fu Z. Effect of cavitation on surface formation mechanism of ultrasonic vibration-assisted EDM. Int J Adv Manuf Technol 2023;124:3645–56. https://doi.org/10.1007/s00170-022-10780-5.

[36] Khan MY, Rao PS, Pabla BS. An experimental study on magnetic field-assisted-EDM process for Inconel-625. Advances in Materials and Processing Technologies 2022;8:2204–30. https://doi.org/10.1080/2374068X.2022.2036450.

[37] Bhowmik A, Kumar R, K RM, Kumar YK, Kumar P, Bains PS, et al. Advancing sustainability in EDM: a brief review of eco-friendly dielectric fluids. AIP Adv 2025;15:040702. https://doi.org/10.1063/5.0269290.

[38] Sai Ram J, Jeavudeen S, Mouda PA, Ahamed N. The role of various dielectrics used in EDM process and their environmental, health, and safety issues. Mater Today Proc 2026;119:192–200. https://doi.org/10.1016/j.matpr.2023.05.137.

[39] Mishra H, Modi YK, Tiwari SK, Maity K. Experimental investigation of micro-EDM hole drilling on Inconel-718 for predicting surface quality characteristics. Journal of Micromanufacturing 2026;9:80–9. https://doi.org/10.1177/25165984251328100.

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[40] Sahoo R, Singh NK, Bajpai V. A novel approach for modeling MRR in EDM process using utilized discharge energy. Mech Syst Signal Process 2023;185:109811. https://doi.org/10.1016/j.ymssp.2022.109811.
[41] Gao R, Loh YM, Li K, Chen R, Jiang C, Cheung CF, et al. Study on the material removal mechanism of cemented carbide magnetic materials in magnetic field-assisted mass polishing. Tribol Int 2025;208:110647. https://doi.org/10.1016/j.triboint.2025.110647.
[42] Rafaqat M, Mufti NA, Ahmed N, Alahmari AM, Hussain A. EDM of D2 Steel: performance Comparison of EDM Die Sinking Electrode Designs. Appl Sci 2020;10:7411. https://doi.org/10.3390/app10217411.
[43] Riaz AA, Ullah N, Hussain G, Alkahtani M, Khan MN, Khan S. Experimental investigations on the effects of rotational speed on temperature and microstructure variations in incremental forming of T6- tempered and Annealed AA2219 aerospace alloy. Metals 2020;10:809. https://doi.org/10.3390/met10060809.
[44] Ma H, Mao W, Zhao N, Zhu H, Wang P, Sun Q, et al. Mechanical properties and microstructure evolution of 2219 aluminum alloy via electromagnetic ring expansion & electromagnetic treatment. J Alloys Compd 2023;947:169615. https://doi.org/10.1016/j.jallcom.2023.169615.
[45] Şimşek Ü, Çoğun C, Esen Z. Effects of electrolytic copper and copper alloy electrodes on machining performance in electrical discharge machining (EDM). Mach Sci Technol 2022;26:229–44. https://doi.org/10.1080/10910344.2022.2044855.
[46] Mao X, Almeida S, Mo J, Ding S. The state of the art of electrical discharge drilling: a review. Int J Adv Manuf Technol 2022;121:2947–69. https://doi.org/10.1007/s00170-022-09549-7.
[47] Prakash T, Ranjith R, Krishna Mohan S, Venkatesan S. Electric discharge machining of AZ91 magnesium hybrid composites under different dielectric mediums. Adv Mater Sci Eng 2022;2022:1–15. https://doi.org/10.1155/2022/3502383.
[48] Ali S, Aidossova A, Begassilova N, Gissa A, Talamona D, Perveen A. Investigating die-sinking EDM drilling performance on additively manufactured SS316L steel lattice structures. Manufacturing Letters 2025;44:651–60. https://doi.org/10.1016/j.mfglet.2025.06.076.

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